Chapter 2: Elementary Probability Theory

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2.1 Introduction

Probability theory is the language of uncertainty. It is through the mathematical treatment of probability theory that we attempt to understand, systematize and thus eventually predict the governance of chance events. The role of probability theory in modeling real life phenomenon, most of which are governed by chance, is somewhat akin to the role of calculus in deterministic physical sciences and engineering. Thus though the study of probability theory is important and interesting in its own right with its applications spanning over fields as diverse as astronomy to zoology, our main interest in probability theory lies in its applicability as a model for distribution of possible values of variables of interest in a population.

We are eventually interested in data analysis, with the data treated as a limited sample, from which we would like to extrapolate or generalize and draw inference about different phenomena of interest in an underlying real or hypothetical population. But in order to do so, we have to first provide a structure in the population of values itself, from which the observed data is but a sample. Probability theory helps us provide this structure. By providing this structure we mean, it enables one to define and thus meaningfully talk about concepts, which are very well-defined in an observed sample like its mean, median, distribution etc., in the population. Without this well-defined population structure, statistical analysis or statistical inference does not have any meaning, and thus these initial notes on probability theory should be regarded as a pre-requisite knowledge for the statistical theory and applications developed in the subsequent notes on mathematical and applied statistics. However the probability concepts discussed here would also be useful for other areas of interest like operations research or systems.

Though our ultimate goal is statistical inference and the role of probability theory in that is loosely as stated above, there are at least two different philosophies which guide this inference procedure. The difference between these two philosophies stems from the very meaning and interpretation of the probability itself. In these notes, we shall generally adhere to the frequentist interpretation of probability theory and its consequence - the so-called classical statistical inference. However before launching on to the mathematical development of probability theory, it would be instructive to first briefly indulge in its different meanings and interpretations.

2.2 Interpretation of Probability

There are essentially three types of interpretations of probabilities, namely,

1. Frequentist Interpretation

2. Subjective Interpretation &
3. Logical Interpretation

2.2.1 Frequentist Interpretation

This is the most standard and conventional interpretation of probability. Consider an experiment, like tossing a coin or rolling a dice, whose outcome cannot be exactly predicted before hand, and which is repeatable. We shall call such an experiment a **chance experiment**. Now consider an event, which is nothing but a statement regarding the outcome of a chance experiment. Like for example the event might be "the result of the coin toss is Head" or "the roll of the dice resulted in an even number". Since the outcome of such an experiment is uncertain, so is the occurrence of an event. Thus we would like to talk about the probability of occurrence of such an event of interest.

In the frequentist sense, probability of an event or outcome is interpreted as its long-term relative frequency over an infinite number of trials of the underlying chance experiment. Note that in this interpretation the basic premise is that the chance experiment under consideration is repeatable. If A is an event for this repeatable chance experiment, then the frequentist interpretation of the statement Probability(A)=p is as follows. Perform or repeat the experiment some n times. Then

$$p = \lim_{n\to\infty} \frac{\text{\# of times the event } A \text{ has occurred in these } n \text{ trials}}{n}$$

Note that since relative frequency is a number between 0 and 1, in this interpretation, so would be the frequentist probability. Also note that since sum of the relative frequencies of two disjoint events A and B (two events A and B are called disjoint if they cannot happen simultaneously) is the relative frequency of the event A OR B, in this interpretation, probability of the event that at least one of the two disjoint events A and B has occurred is same as the sum of their individual probabilities.

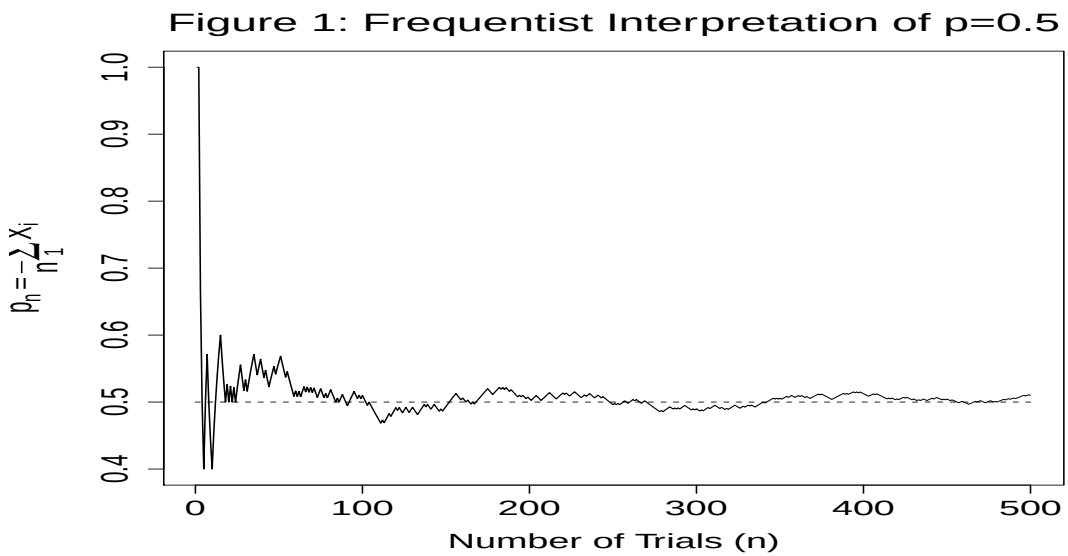
Now coming back to the numerical interpretation in the frequentist sense, as a concrete example, consider the coin tossing experiment and the event of interest "the result of the coin toss is Head". Now how can a statement like "probability of getting a Head in a toss of this coin is 0.5" be interpreted in frequentist terms? (Note that by the aforementioned remark, probability, being a relative frequency has to be a number between 0 and 1.) The answer is as follows. Toss the coin n times. For the i-th toss let

$$X_i = \begin{cases} 1 & \text{if the } i\text{-th toss resulted in a Head} \\ 0 & \text{otherwise} \end{cases}.$$

Now keep track of the relative frequency of Head till the n-th toss, which is given by

$$\hat{p}_n = \frac{1}{n}\sum_{i=1}^{n} X_i.$$

Then according to the frequentist interpretation, probability of getting a Head is 0.5 means $\hat{p}_n \to 0.5$ as $n \to \infty$. This is illustrated in Figure 1. 500 tosses of a fair coin was simulated by a computer and the resulting $\hat{p}_n$'s were plotted against n for $n = 1, 2, \ldots, 500$. The dashed line in Figure 1 has the equation $\hat{p}_n = 0.5$. Observe how the $\hat{p}_n$'s are converging to this value as n is getting larger. This is the underlying frequentist interpretation of "probability of getting a Head in a toss of a coin is 0.5".



2.2.2 Subjective Interpretation

While the frequentist interpretation works fine for a large number of cases, its major drawback is this interpretation requires the underlying chance experiment to be repeatable, which need not necessarily always be the case. Experiments like tossing a coin, rolling a dice, drawing a card, observing heights, weights, ages, incomes of individuals etc. are repeatable and thus probabilities of events associated with such experiments can very comfortably be interpreted as their long-term relative frequencies.

But what about probabilities of events like, "it will rain tonight" or "the new venture capital company X will go bust within a year" or "Y will not show up on time for the movie"? None of these events are repeatable in the sense that they are just one-time phenomenon. It will either rain tonight or it won't, company X will either go bust within a year or it won't, Y will either show up for the movie on time or she won't. There is no scope of observing a repeated trial of tonight's performance w.r.t. rain, or no scope of observing repeated performance of company X during the first year of its inception, or no scope of repeating an identical situation for someone waiting for Y in front of the movie-hall.

All the above events pertain to non-repeatable one-time phenomena. Yet since the outcomes of these phenomena are uncertain, it is only but natural for us to attempt to quantify these uncertainties in terms of probabilities. Indeed most of our everyday personal experiences with uncertainties involve such one-time phenomenon (Shall I get this job? Shall I be able

to reach the airport on time? Will she go out with me for dinner?), and we usually either consciously or unconsciously attach some probabilities with them. The exact numbers we attach to these probabilities most of the time are not very clear in our mind, and we shall shortly describe an easy method to do so, but the point is that such numbers are necessarily personal or subjective in nature. You might feel the probability that it will rain tonight is 0.6, while in my assessment the probability of the same event might be 0.5, while your friend might think that this probability is 0.4. Thus for the same event different persons might assess its chance differently in their mind giving rise to different subjective or personal probabilities for the same event. This is an alternative interpretation of probability.

Now let us discuss a simple method of how to elicit a precise number between 0 and 1 as a subjective probability one is associating with a particular (possibly one-time) event E. To be concrete let E be the event. "it will rain tonight". Now consider a betting scheme on the occurrence of the event E, which says that you will get Rs.1 if the event E occurs, and will get nothing if it does not occur. Since you have some chance of winning that Rs.1 (think of it as a lottery) without any loss to you (in the worst case scenario of non-occurrence of E you do not get anything) it is only but fair to ask you to pay some entry fee to get into this bet. Now what in your mind is a "fair" entry fee for this bet? If you feel that Rs.0.50 is a "fair" entry fee for getting into this bet, then in your mind you are thinking that it is equally likely that it will rain as it will not rain, and thus the subjective probability you are associating with E is 0.5. But on the other hand suppose you are thinking that it is more likely that it will rain tonight than it will not. Then since in your mind you are thinking that you are more likely to win that Rs.1 than nothing, you must consider something more than Rs.0.50 as a "fair" entry fee. Actually in this case anything less than Rs.0.50 would be a "fair" price to you, since in your judgment it is more likely to rain than it is not, you would stand to gain if you pay anything less than Rs.0.50 as entry fee to enter into the bet. So think of the "fair" entry fee as that amount which is the maximum you are willing to pay to get into this bet. Now what is this maximum amount you are willing to shell out as the entry-fee, so that you consider the bet to be still "fair"? Is it Rs.0.60? Then your subjective probability of E is 0.6. Is it Rs.0.82? Then your subjective probability of E is 0.82. Similarly if you think that it is more likely that it will not rain tonight than it will, you will not consider an entry fee of more than Rs.0.50 to be "fair". It has to be something less than Rs.0.50. But how much? Will you enter the bet for Rs.0.40 as the entry fee? If yes, then in your mind the subjective probability of E is 0.4. If you still consider Rs.0.40 to be too high a price for this bet then come down further and see at what price you are willing to get into the bet. If to you the fair price is Rs.0.13 then your subjective probability of E is 0.13.

Interestingly even with a subjective interpretation of probability, in terms of an entry fee for a "fair" bet, by its very construction it becomes a number between 0 and 1. Furthermore it may be shown that such subjective probabilities are also required to follow the standard probability laws. Proofs of subjective probabilities abiding by these laws are provided in Appendix B of my notes on "Bayesian Statistics" and the interested reader is encouraged to go through it after finishing this chapter.

2.2.3 Logical Interpretation

A third view of probability is that it is the mathematics of inductive logic. By this we mean that as the laws of Boolean Algebra govern Aristotelean deductive logic, similarly the probability laws govern the rules of inductive logic. Deductive logic is essentially founded on the following two basic syllogisms:

D.Syllogism 1. If A is true then B is true. A is true, therefore B must be true.

D.Syllogism 2. If A is true then B is true. B is false, therefore A must be false.

Inductive logic tries to infer from the other side of the implication sign and beyond, which may be summarized as follows:

I.Syllogism 1. If A is true then B is true. B is true, therefore A becomes "more likely" to be true.

I.Syllogism 2. If A is true then B is true. A is false, therefore B becomes "more likely" to be false.

I.Syllogism 3. If A is true then B is "more likely" to be true. B is true, therefore A becomes "more likely" to be true.

I.Syllogism 4. If A is true then B is "more likely" to be true. A is false, therefore B becomes "more likely" to be false.

Starting with a set of minimal basic desiderata, which qualitatively state what "more likely" should mean to a rational being, one can show after some mathematical derivation that it is nothing but a notion which must abide by the laws of probability theory, namely the complementation law, addition law and multiplication law. Starting from the mathematical definition of probability, irrespective of its interpretation, these laws have been derived in §5. Thus for readers unfamiliar with these laws, it would be better to come back to this sub-section after §5, because these laws would be needed to appreciate how probability may be interpreted as inductive logic, as stated in the I.Syllogisms above.

Let "If A is true then B is true" be true, and $P(X)$ and $P(X^c)$ respectively denote the chances of X being true and false, and $P(X|Y)$ denote the chance of X being true when Y is true, where X and Y are placeholders for A, B A^c or B^c. Then I.Syllogism 1 claims that $P(A|B) \geq P(A)$. But since $P(A|B) = P(A)\frac{P(B|A)}{P(B)}$, $P(B|A) = 1$ and $P(B) \leq 1$, $P(A|B) \geq P(A)$. Similarly I.Syllogism 2 claims that $P(B|A^c) \leq P(B)$. This is true because $P(B|A^c) = P(B)\frac{P(A^c|B)}{P(A^c)}$ and by I.Syllogism 1 $P(A^c|B) \leq P(A^c)$. The premise of I.Syllogisms 3 and 4 is $P(B|A) \geq P(B)$ which implies $P(A|B) = P(A)\frac{P(B|A)}{P(B)} \geq P(A)$ proving I.Syllogism 3. Similarly since by I.Syllogism 3 $P(A^c|B) \leq P(A^c)$ and $P(B|A^c) = P(B)\frac{P(A^c|B)}{P(A^c)}$, $P(B|A^c) \leq P(B)$ proving I.Syllogism 4.

As a matter of fact D.Syllogisms 1 and 2 also follow from the probability laws. The claim of D.Syllogism 1 is that $P(B|A) = 1$, which follows from the observation that $P(A\&B) = P(A)$ (because of the fact that, If A is true then B is true) and $P(B|A) = P(A\&B)/P(A) = 1$.

Similarly $P(A|B^c) = P(A\&B^c)/P(B^c) = 0$, since the chance of A being true and simultaneously B being false is 0, proving D.Syllogism 2. This shows probability as an extension of deductive logic to inductive logic which yields deductive logic as a special case.

Logical interpretation of probability may be thought of as a combination of both objective and subjective approaches. In this interpretation numerical values of probabilities are necessarily subjective. By that it is meant that probability must not be thought of as an intrinsic physical property of the phenomenon, it should rather be viewed as the degree of belief about the truth of a proposition by an observer. Pure subjectivists hold that this degree of belief might differ from observer to observer. Frequentists hold it as a pure objective quantity independent of the observer like mass or length which may be verified by repeated experimentation and calculation of relative frequencies. In its logical interpretation, though probability is subjective, in the sense that it is not a physical quantity which is intrinsic to the phenomenon and it only resides in the observer's mind, it is also an objective number, in the sense that no matter who the observer is, given the same set of information and the state of knowledge, each rational observer must assign the same probabilities. A coherent theory of this logical approach shows not only how to assign these initial probabilities, it goes on to show how to assimilate knowledge in terms of observed data, and systematically carry out this induction about uncertain events, and thus providing a solution to problems which are in general regarded as statistical in nature.

2.3 Basic Terminologies

Before presenting the probability laws, as has been referred to from time to time in §2, it would be useful to first systematically introduce the basic terminologies and their mathematical definitions including that of probability. In this discussion we shall mostly confine ourselves in repeatable chance experiments. This is because 1) our focus here is frequentist in nature, and 2) the exposition is easier. It is because of the second reason that most standard probability texts also adhere to the frequentist approach while introducing the subject. Though familiarity with the frequentist treatment is not a pre-requisite, understanding the development of probability theory from the subjective or logical angle becomes a little easier for the reader already acquainted with the basics from a "standard" frequentist perspective. We start our discussion by first providing some examples of repeatable chance experiments and chance events.

Example 2.1 **A:** Tossing a coin once. This is a chance experiment because you cannot predict the outcome of this experiment, which will be either a Head (H) or Tail (T), beforehand. For the same reason, the event, "the result of the toss is Head", is a chance event.

B: Rolling a dice once. This is a chance experiment because you cannot predict the outcome of this experiment, which will be one of the integers 1, 2, 3, 4, 5, or 6, beforehand. Likewise the event, "the outcome of the roll is an even number", is a chance event.

C: Drawing a card at random from a deck of standard playing card is a chance experiment and "the card drawn is Ace of Spade" is a chance event.

D: Observing the number of weekly accidents in a factory is a chance experiment and "no accident has occurred this week" is a chance event.

E: Observing how long a light bulb lasts is a chance experiment and "the bulb lasted for more than a 1000 hours" is a chance event. $\bigtriangledown$

As in the above examples, the systematic study of any chance experiment starts with the consideration of all possibilities that can occur. This leads to our first definition.

Definition 2.1: The set of all possible outcomes of a chance experiment is called the **sample space** and is denoted by Ω. A simple single outcome is denoted by ω.

Example 2.1 (Continued) **A:** For the chance experiment - tossing a coin once, $\Omega = \{H, T\}$.

B: For the chance experiment - rolling a dice once, $\Omega = \{1, 2, 3, 4, 5, 6\}$.

C: For the chance experiment - drawing a card at random from a deck of standard playing cards, $\Omega = \{\clubsuit 2, \clubsuit 3, \ldots, \clubsuit K, \clubsuit A, \diamondsuit 2, \diamondsuit 3, \ldots, \diamondsuit K, \diamondsuit A, \heartsuit 2, \heartsuit 3, \ldots, \heartsuit K, \heartsuit A, \spadesuit 2, \spadesuit 3, \ldots, \spadesuit K, \spadesuit A\}$.

D: For the chance experiment - observing the number of weekly accidents in a factory, $\Omega = \{0, 1, 2, 3, \ldots\} = \mathcal{N}$, the set of natural numbers.

E: For the chance experiment - observing how long does a light-bulb last, $\Omega = [0, \infty) = \Re^+$, the non-negative half of the real line $\Re$. $\bigtriangledown$

Example 2.2: **A:** If the experiment is tossing a coin twice, $\Omega = \{HH, HT, TH, TT\}$.

B: If the experiment is rolling a dice twice, $\Omega = \{(1,1), \ldots, (1,6), \ldots, \ldots, (6,1), (6,6)\} = \{\text{ordered pairs } (i,j) : 1 \leq i \leq 6, 1 \leq j \leq 6, i \text{ and } j \text{ integers}\}$. $\bigtriangledown$

We have so far been loosely using the term "event". In all practical applications of probability theory the term "event" may be used as in everyday language, namely, a statement or proposition about some feature of the outcome of a chance experiment. However to proceed further it would be necessary to give this term a precise mathematical meaning.

Definition 2.2: An **event** is a subset of the sample space. We typically use upper-case Roman alphabets like A, B, E etc. to denote an event. [1]

[1] Strictly speaking this definition is not correct. For a mathematically rigorous treatment of probability theory it is necessary to confine oneself only to a collection of subsets of Ω, and not all possible subsets. Only members of such a collection of subsets of Ω will qualify to be called as an event. As shall be seen shortly, since we shall be interested in set-theoretic operations with the events and their results, such a collection of subsets of Ω, to be able to qualify as a collection of events of interest, must satisfy some non-emptiness and closure properties under set-theoretic operations. In particular a collection of events $\mathcal{A}$, consisting of subsets of Ω must satisfy

i. $\Omega \in \mathcal{A}$, ensuring that the collection $\mathcal{A}$ is non-empty.

ii. $A \in \mathcal{A} \Longrightarrow A^c = \Omega - A \in \mathcal{A}$, ensuring the collection $\mathcal{A}$ is closed under the complementation operation.

iii. $A_1, A_2, \ldots \in \mathcal{A} \Longrightarrow \bigcup_{n=1}^{\infty} A_n \in \mathcal{A}$, ensuring that the collection $\mathcal{A}$ is closed under countable union operation.

A collection $\mathcal{A}$ satisfying the above three properties is called a $\sigma-$field, and the collection of all possible events is required to be a $\sigma-$field. Thus in rigorous mathematical treatment of the subject it is not enough

As mentioned in the paragraph immediately preceding Definition 2, typically an event would be a linguistic statement regarding the outcome of a chance experiment. It will then usually be the case that this statement then can be equivalently expressed as a subset E of Ω, meaning the event (as understood in terms of the linguistic statement) would have occurred if and only if the outcome is one of the elements of the set $E \subseteq \Omega$. On the other hand, given a subset A of Ω, it is usually the case that one can express the commonalities of the elements of A in words, and thus construct a linguistic statement equivalent to the mathematical notion (a subset of Ω) of the event. A few examples will help clarify this point.

Example 2.1 (Continued) A: The event "the result of the toss is Head" mathematically corresponds to $\{H\} \subseteq \{H, T\} = \Omega$, while the null set $\phi \subseteq \Omega$ corresponds to the event "nothing happens as a result of the toss".

B: The event "the outcome of the roll is an even number" mathematically corresponds to $\{2, 4, 6\} \subseteq \{1, 2, 3, 4, 5, 6\} = \Omega$. The set $\{2, 3, 5\}$ corresponds to a drab linguistic description of the event "the outcome of the roll is a 2, or a 3 or a 5" or something a little more interesting like "the outcome of the roll is a prime number". $\bigtriangledown$

Example 2 B (Continued): For the rolling a dice twice experiment the event "the sum of the rolls equals 4" corresponds to the set $\{(1, 3), (2, 2), (3, 1)\}$. $\bigtriangledown$

Example 3: Consider the experiment of tossing a coin three times. Note that this experiment is equivalent to tossing three (distinguishable) coins simultaneously. For this experiment the sample space $\Omega = \{HHH, HHT, HTH, THH, TTH, THT, HTT, TTT\}$. The event "total number of heads in the three tosses is at least 2" corresponds to the set $\{HHH, HHT, HTH, THH\}$. $\bigtriangledown$

Now that we have familiarized ourselves with the systematization of the basics of chance experiments, it is now time to formalize or quantify "chance" itself in terms of probability. As noted in §2, there are different alternative interpretations of probability. It was also pointed out there that no matter what the interpretation might be they all have to follow the same probability laws. In fact in subjective/logical interpretation the probability laws, yet to be proved from the following definition, are derived (with a lot of mathematical details) directly from their respective interpretations, while the same can somewhat obviously be done with the frequentist interpretation. But no matter how one interprets probability, except for a very minor technical difference (countable additivity versus finite additivity for the subjective/logical interpretation) there is no harm in defining probability in the following abstract mathematical way, which is true for all its interpretations. This enables one to study the mathematical theory of probability without getting bogged down with its philosophical meaning, though its development from a purely subjective or logical angle might appear to be somewhat different.

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just to consider the sample space Ω, one must consider the pair $(\Omega, \mathcal{A})$, the sample space Ω and $\mathcal{A}$, a $\sigma-$field of events of interest consisting of subsets of Ω. This consideration stems from the fact that in general it is not possible to assign probabilities to all possible subsets of Ω, and one confines oneself only to those subsets of interest for which one can meaningfully talk about their probabilities. In our quasi-rigorous treatment of probability theory, since we shall not encounter such difficulties, without much harm, we shall pretend as if such pathologies do not arise and for us the collection of events of interest $= \wp(\Omega)$, called the power set of Ω, which consists of all possible subsets of Ω.

Definition 2.3: **Probability** $P(\cdot)$ is a function with subsets of Ω as its domain and real numbers as its range, written as $P : \mathcal{A} \to \Re$, where $\mathcal{A}$ is the collection of events under consideration (which as stated in footnote 1 may be pretended to be equal to $\wp(\Omega)$), such that

i. $P(\Omega) = 1$

ii. $P(A) \geq 0 \; \forall A \in \mathcal{A}$, and

iii. If $A_1, A_2, \ldots$ are mutually exclusive (meaning $A_i \cap A_j = \phi$ for $i \neq j$), $P\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} P(A_n)$.

Sometimes particularly in subjective/logical development, **iii** above, called countable additivity is considered to be too strong or redundant and instead is replaced by finite additivity:

iii'. For A, $B \in \mathcal{A}$ and $A \cap B = \phi \Longrightarrow P(A \cup B) = P(A) + P(B)$.

Note that **iii** $\Rightarrow$ **iii'**, because, for A, $B \in \mathcal{A}$ and $A \cap B = \phi$, let $A_1 = A$, $A_2 = B$ and $A_n = \phi$ for $n \geq 3$. Then by **iii**, $P(A \cup B) = P\left(\bigcup_{n=1}^{\infty} A_n\right) = P(A) + P(B) + \sum_{n=3}^{\infty} P(\phi)$, and for the right hand side to exist $P(\phi)$ must equal 0, implying $P(A \cup B) = P(A) + P(B)$.

Though definition 3 precisely states what numerical values probabilities of two extreme elements *viz.* ϕ and Ω of $\mathcal{A}$ must take, (0 and 1 respectively, that $P(\phi) = 0$ has just been shown, and **i** states $P(\Omega) = 1$) it does not say anything about the probabilities of the intermediate sets. Actually assignment of probabilities to such non-trivial sets is precisely the role of statistics, and the theoretical development of probability as inductive logic leads to a such coherent (alternative Bayesian) theory of statistics. However even otherwise it is still possible to logically argue and develop probability models without resorting to their empirical statistical assessments, and that is precisely what we have set ourselves to do in these notes on probability theory. Indeed empirical statistical assessments of probability in the frequentist paradigm also typically starts with such a logically argued probability model and thus it is imperative that we first familiarize ourselves with such logical probability calculations. Towards this end we begin our initial probability computations for a certain class of chance experiments using the so-called classical or *apriori* method, which are essentially based on combinatorial arguments.

2.4 Combinatorial Probability

Historically probabilities of chance events for experiments like coin tossing, dice rolling, card drawing etc. were first worked out using this method. Thus this method is also known as *classical* method of calculating probability. [2] This method applies only in situations where the sample space Ω is finite. The basic premise of the method is that since we do not have

[2]Though some authors refer to this as one of the interpretations of probability, it is possibly better to view this as a method of calculating probability for a certain class of repeatable chance experiments in the absence of any experimental data, rather than one of the interpretations. The number one gets as a result of such *classical* probability calculation of an event may be interpreted as either its long-term relative frequency, or one's logical belief about it for an *apriori* subjective assignment of a uniform distribution over the set of all possibilities, which may be intuitively justified as, "since I do not have any reason to favor the possibility

any experimental evidence to think otherwise, let us assume *apriori* that all possible (atomic) outcomes of the experiment are *equally likely*[3]. Now suppose the finite Ω has N elements, and an event $E \subseteq \Omega$ has $n \leq N$ elements. Then by (finite) additivity, probability of E equals n/N. In words, probability of an event E,

$$P(E) = \frac{\text{\# of outcomes favorable to the event } E}{\text{Total number of possible outcomes}} = \frac{n}{N} \tag{1}$$

Example 2.4: A machine contains a large number of screws. But the screws are only of three sizes small (S), medium (M) and large (L). An inspector finds 2 of the screws in the machine are missing. If the inspector carries only one screw each of each size, the probability that he will be able to fix the machine then and there is 2/3. The sample space of possibilities for the two missing screws is Ω = {SS, SM, SL, MS, MM, ML, LS, LM, LL} which has 9 elements. Out of these if the missing screws were Ω-{SS,MM,LL} the inspector could fix the machine then and there. Since this event has 6 elements, the probability of this event is 2/3.

Example 2.2 B (Continued): Rolling a "fair"[4] dice twice. This experiment has 36 equally likely fundamental outcomes. Thus since the event "the sum of the rolls equals 4" contains just 3 of them, its probability is 1/12. Likewise the event "one of the rolls is at least 4" = $\{(4,1),\ldots,(4,6),\ \ (5,1),\ldots(5,6),\ \ (6,1),\ldots,(6,6),\ \ (1,4),(2,4),(3,4),\ \ (1,5),(2,5),(3,5),$ $(1,6),(2,6),(3,6)\}$, having $3\times 6 + 3\times 3 = 27$ outcomes favorable to it, has probability 3/4.

In the above examples though we have attempted to explicitly write down the sample space Ω and the sets corresponding to the events of interest, it should also be clear from these examples that such explicit representations are strictly not required for the computation of classical probabilities. What is important is only the number of elements in them. Thus in order to be able to compute classical probabilities, we must first learn to systemically count. We first describe the fundamental counting principle, and then go on developing different counting formulæ, which are frequently encountered in practice. All these commonly occurring counting formulæ are based on the fundamental counting principle. We provide separate formulæ for them so that one need not reinvent the wheel every time one encounters such standard cases. However it should be borne in mind that though quite extensive, the array of counting formulæ provided here are by no means exhaustive and it is impossible to provide such a list. Very frequently situations will arise where no standard formula, such as the ones described here, will apply and in those situations counting needs to be done by developing new formula by falling back upon the fundamental counting principle.

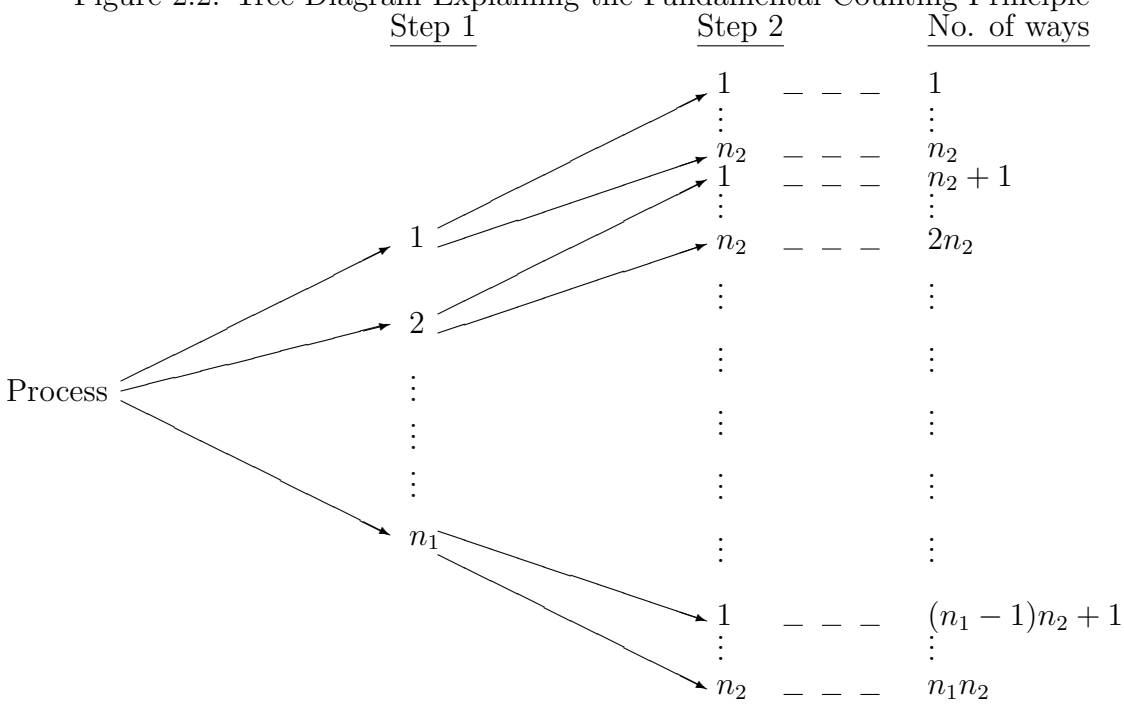
Fundamental Counting Principle: If a process is accomplished in two steps with n_1 ways to do the first step and n_2 ways to do the second, then the process is accomplished totally in $n_1 n_2$ ways. This is because each of the n_1 ways of doing the first step is associated with each of the n_2 ways of doing the second step. This reasoning is further clarified in Figure 2.

of one outcome over the other, it is but natural for me to assume *apriori* that all of them have the same chance of occurrence".

[3]This is one of the fundamental criticisms of classical probability, because it is defining probability in its own terms and thus leading to a circular definition.

[4]Now we qualify the dice as fair, for justifying the equiprobable fundamental outcomes assumption, the pre-requisite for classical probability calculation.

Figure 2.2: Tree Diagram Explaining the Fundamental Counting Principle



Like for example if you have 10 tops and 8 trousers you can dress in 80 different ways. Repeating the principle twice, if a restaurant offers one a choice of one item each from its menu of 8 appetizers, 6 éntrees and 4 desserts for a full dinner, one can construct 192 different dinner combinations. If customers are classified according to 2 genders, 3 marital status (never-married, married, divorced/widowed/separated), 4 eduction levels (illiterate, school drop-out, school certificate only and college graduates), 5 age groups ($<$ 18,18-25,25-35,35-50, and 50+) and 6 income levels (very poor, poor, lower-middle class, middle-middle-class, upper-middle-class and rich) then repeated application of the principle yields 720 distinct demographic groupings.

Starting with the above counting principle one can now now develop many useful standard counting methods, which are summarized below. But before that let us first introduce the factorial notation. For a positive integer n, $n!$ (read as "factorial n") $= 1.2.\ldots(n-1).n$. Thus 1!=1, 2!=2, 3!=6, 4!=24, 5!=120 etc. 0! is defined to be 1.

Some Counting Formulæ:

Formula 1. The number of ways in which k distinguishable balls (say either numbered or say of different colors) can be placed in n distinguishable cells equals n^k. This is because the first ball may be placed in n ways in any one of the n cells. The second ball may again be placed in n ways in any one of the n cells, and thus the number of ways one can place the first two balls equals $n \times n = n^2$, according to the fundamental counting principle. Reasoning in this manner it may be seen that the number of ways the k balls may be placed in n cells equals $\underbrace{n \times n \times \cdots \times n}_{k\text{-times}} = n^k$. $\bigtriangledown$

Example 2.5: The probability of obtaining at least one ace in 4 rolls of a fair dice equals $1 - (5^4/6^4)$. To see this first note that it is easier to compute the probability of the comple-

mentary event and then compute the probability of the event of interest by subtracting the probability of the complementary event from 1, following the complementation law (*vide.* §5). Now complement of the event of interest "at least one ace in 4 rolls " is "no ace in 4 rolls". Total number of possible outcomes of 4 rolls of a dice equals $6 \times 6 \times 6 \times 6 = 6^4$ (each roll is a ball which can fall in any one of the 6 cells). Similarly the number of outcomes favorable to the event "no ace in 4 rolls" equals 5^4 (for any given roll it not ending up with an ace means it has rolled into either a 2, 3, 4, 5 or 6 - 5 possibilities). Thus by (1) the probability of the event "no ace in 4 rolls" equals $5^4/6^4$, and by complementation law, the probability of the event "at least one ace in 4 rolls " equals $1 - (5^4/6^4)$. $\bigtriangledown$

Example 2.6: In an office with the usual 5 days week, which allows its employees 12 casual leaves in a year, the probability that all the casual leaves taken by Mr. X last year were either a Friday or a Monday equals $2^{12}/5^{12}$. The total number of possible ways in which Mr. X could have taken his 12 casual leaves last year equals 5^{12}, (each of the last year's 12 casual leaves of Mr. X is a ball which could have fallen on one of the 5 working days as cells) while the number of ways in which the 12 casual leaves could have been taken on either a Friday or a Monday equals 2^{12}. Thus the sought probability equals $2^{12}/5^{12} = 1.677 \times 10^{-5}$ which is extremely slim. Thus we cannot possibly blame Mr X's boss if she is suspecting him of using his casual leaves for enjoying extended long weekends! $\bigtriangledown$

Formula 2. The number of possible ways in which k objects drawn without replacement from n distinguishable objects ($k < n$) can be arranged between themselves is called the number of **permutations** of k out of n. This number is denoted by ${}^nP_k$ or $(n)_k$ (read as "n-P-k") and equals $n!/(n-k)!$. We shall draw the objects one by one and then place them in their designated positions like the first position, second position, ... , k-th position to get the number of all possible arrangements. The first position can be filled in n ways. After filling the first position (since we are drawing objects without replacement) there are $n-1$ objects left and hence the second position can be filled in $n-1$ ways. Therefore according to the fundamental counting principle the number of possible arrangements for filling the first two positions equals $n \times (n-1)$. Proceeding in this manner when it comes to fill the k-th position we are left with $n-(k-1)$ objects to choose from, and thus the total number of possible arrangements of k objects taken from an original set of n objects equals $n.(n-1)\ldots(n-k+2).(n-k+1) = \frac{n.(n-1)\ldots(n-k+2).(n-k+1).(n-k).(n-k-1)\ldots 2.1}{(n-k).(n-k-1)\ldots 2.1} = n!/(n-k)!$. $\bigtriangledown$

Example 2.7: An elevator starts with 4 people and stops at each of the 6 floors above it. The probability that everybody gets off at different floors equals $(6)_4/6^4$. The total number of possible ways in which the 4 people can disembark the elevator equals 6^4 (each person is a ball and each floor is a cell). Now the number of cases where everybody disembarks at different floors is same as choosing 4 distinct floors from the available 6 for the four different people and then taking their all possible arrangements, which can be done in $(6)_4$ ways, and thus the required probability equals $(6)_4/6^4$. $\bigtriangledown$

Example 2.8: The probability that in a group of 8 people birthdays of at least two people will be in the same month is 95.36%. As in example 5, here it is easier to first calculate the probability of the complementary event. The complementary event says that birthdays of all the 8 persons are in different months. The number of ways that can happen is same as choosing 8 months from the total of possible 12 and then considering their all possible

arrangements, which can be done in $(12)_8$ ways. Now the total number of possibilities for the months of birthdays of 8 people is same as the number of possibilities of placing 8 balls in 12 cells, which equals 12^8. Hence the probability of the event "no two person's birthdays are in the same month" is $(12)_8/12^8$, and by the complementation law (*vide.* §5), the probability that at least two person's birthdays are in the same month equals 1-$(12)_8/12^8$=0.9536. $\bigtriangledown$

Example 2.9: Given n keys and only one of which will open a door, the probability that the door will open in the k-th trial, $k = 1, 2, \ldots, n$, where the keys are being tried out one after another till the door opens, does not depend on k and equals $1/n$ $\forall k = 1, 2, \ldots, n$. The total number of possible ways in which the trial can go up to the k-th try is same as choosing k out of the n keys and trying them in all possible orders which is given by $(n)_k$. Now among these possibilities the number of cases where the door does not open in the first $(k-1)$ tries and then opens in the k-th trial is the number of ways one can try $(k-1)$ "wrong" keys from the total set of $(n-1)$ wrong keys in all possible order, which can be done in $(n-1)_{k-1}$ ways. Thus the required probability $= \frac{(n-1)_{k-1}}{(n)_k} = \frac{(n-1).(n-2)...\{(n-1)-(k-3)\}.\{(n-1)-(k-2)\}}{n.(n-1)...(n-k+2).(n-k+1)} = \frac{1}{n}$. $\bigtriangledown$

Formula 3. The number of ways one can choose k objects from a set of n distinguishable objects just to form a group without bothering about the order in which the objects appeared in the selected group is called the number of **combinations** of k out of n. This number is denoted by ${}^nC_k$ (read as "n-C-k") or $\begin{pmatrix} n \\ k \end{pmatrix}$ (read as "n-choose-k") and equals $\frac{n!}{k!(n-k)!}$. First note that the number of possible arrangements one can make by drawing k objects from n is already given by $(n)_k$. Here we are concerned about the possible number of such groups without bothering about the arrangements of the objects within the group. That is as long as the group contains the same elements it is the counted as one single group irrespective of the order in which the objects are drawn or arranged. Now among the $(n)_k$ possible permutations there are arrangements which consist of basically the same elements but they are counted as distinct because the elements appear in different order. Thus if we can figure out how many such distinct arrangements of the same k elements are there, then all these will represent the same group. Since these were counted as different in the $(n)_k$ many permutations, dividing $(n)_k$ by this number will give $\begin{pmatrix} n \\ k \end{pmatrix}$ or the total number of possible groups of size k that can be chosen out of n objects. k objects can be arranged between themselves in $(k)_k = k!/0! = k!$ ways. Hence $\begin{pmatrix} n \\ k \end{pmatrix} = (n)_k/k! = \frac{n!}{k!(n-k)!}$. $\bigtriangledown$

Example 2.10: A box contains 20 screws 5 of which are defective (improperly grooved). The probability that in a random sample of 10 such screws none are defective equals $\begin{pmatrix} 15 \\ 10 \end{pmatrix} \Big/ \begin{pmatrix} 20 \\ 10 \end{pmatrix}$. This is because the total number of ways in which 10 screws can be drawn out of 20 screws is $\begin{pmatrix} 20 \\ 10 \end{pmatrix}$, while the event of interest can happen if and only if all the 10 screws are chosen from the 15 good ones, which can be done in $\begin{pmatrix} 15 \\ 10 \end{pmatrix}$ ways. The prob-

ability of the event "exactly 2 defective screws" in this same experiment is $\dfrac{\binom{15}{8}\binom{5}{2}}{\binom{20}{10}}$.

This is because here the denominator remains same as before, but now the event of interest can happen if and only if one chooses 8 good screws and 2 defective ones. 8 good screws must come from the 15, which can be chosen in $\binom{15}{8}$ ways, while the 2 defective ones must come from the 5, which can be chosen in $\binom{5}{2}$ ways. Now each way of choosing the 8 good ones is associated with each way of choosing the 2 defective ones and thus by fundamental counting principle the number of outcomes favorable to the event "exactly 2 defective screws" equals $\binom{15}{8}\binom{5}{2}$. $\triangledown$

Example 2.11: A group of $2n$ boys and $2n$ girls are randomly divided into groups of equal size. The probability that each group contains an equal number of boys and girls equals $\binom{2n}{n}^2 \Big/ \binom{4n}{2n}$. This is because the number of ways in which a total of $4n$ individuals ($2n$ boys + $2n$ girls) can be divided in two groups of equal size is same as choosing half of these individuals, which equals $2n$, from the original set of $4n$, which can be done in $\binom{4n}{2n}$ ways. Now each of these two groups will have equal number of boys and girls if and only if each group contains n boys and n girls each. Thus the number of outcomes favorable to the event must equal the total number of ways in which we can choose n boys from a total of $2n$ and n girls from a total of $2n$, each of which can be done in $\binom{2n}{n}$ ways, and thus the numerator must equal $\binom{2n}{n}^2$. $\triangledown$

Example 2.12: A man parks his car in a parking lot with n slots in a row in one of the middle slots *i.e.* not at either end. Upon his return he finds that there are now m ($< n$) cars parked in the parking lot, including his own. We want to find the probability of the owner finding both the slots adjacent to his car being empty. The number of ways in which the remaining $m-1$ cars (excluding his own) can occupy the remaining $n-1$ slots equals $\binom{n-1}{m-1}$. Now if both the slots adjacent to the owner's car are empty, the remaining $m-1$ cars must be occupying the slots from among the available $n-3$, which can happen in $\binom{n-3}{m-1}$ ways. Thus the required probability is $\binom{n-3}{m-1} \Big/ \binom{n-1}{m-1}$. $\triangledown$

Formula 4. The combination formula $\binom{n}{k}$ arises from the consideration, the number of groups of size k one can form by drawing objects (without replacement) from a parent set of n distinguishable objects. Because of their appearance in the expansion of the binomial ex-

pression $(a+b)^n$, $\begin{pmatrix} n \\ k \end{pmatrix}$'s are called binomial coefficients. Likewise the coefficients appearing in the expansion of the multinomial expression $(a_1 + a_2 + \cdots + a_k)^n$ are called **multinomial coefficients** with a typical multinomial coefficient denoted by $\begin{pmatrix} n \\ n_1, n_2, \ldots, n_k \end{pmatrix}$ (read as "n-choose-n_1, n_2 etc. n_k") which equals $\frac{n!}{n_1!n_2!\ldots n_k!}$ for $\sum_{i=1}^{k} n_i = n$. The combinatorial interpretation of the multinomial coefficients is, the number of ways one can divide n objects into k ordered groups[5] with the i-th group containing n_i objects $i = 1, 2, \ldots, k$. This is because there are $\begin{pmatrix} n \\ n_1 \end{pmatrix}$ ways of choosing the elements of the first group, then there are $\begin{pmatrix} n - n_1 \\ n_2 \end{pmatrix}$ ways of choosing the elements of the second group and so on, and finally there are $\begin{pmatrix} n - n_1 - \cdots - n_{k-1} \\ n_k \end{pmatrix}$ ways of choosing the elements of the k-th group. So the total number of possible ordered groups equals $\begin{pmatrix} n \\ n_1 \end{pmatrix}\begin{pmatrix} n - n_1 \\ n_2 \end{pmatrix} \cdots \begin{pmatrix} n - n_1 - \cdots - n_{k-1} \\ n_k \end{pmatrix}$ $= \frac{n!}{n_1!(n-n_1)!}\frac{(n-n_1)!}{n_2!(n-n_1-n_2)!} \cdots \frac{(n-n_1-\cdots-n_{k-1})!}{n_k!0!} = \frac{n!}{n_1!n_2!\ldots n_k!}$.

An alternative combinatorial interpretation of the multinomial coefficient is the number of ways one can permute n objects, consisting of k types where for $i = 1, 2, \ldots, k$, the i-th type contains n_i identical copies of those objects which are indistinguishable among themselves. This is because n distinct objects (one object each of each type) can be permuted in $n!$ ways. Now since n_1 of them are identical or indistinguishable, all possible permutations of these n_1 objects among themselves with the other objects fixed in their place will yield the same permutation in this case, which were counted as different in the $n!$ permutations of distinct objects. Now how many such permutations of n_1 objects among themselves are there? There are $n_1!$ such. So with the other objects fixed and regarded as distinct and taking care of indistinguishability of the n_1 objects, the number of possible permutations are $n/n_1!$. Reasoning in the same fashion for the remaining $k-1$ types of objects now it may be seen that the number of possible permutations of n objects with n_i identical copies of the i-th type for $i = 1, 2, \ldots, k$, equals $\frac{n!}{n_1!n_2!\ldots n_k!}$. Thus for example one can form $5! = 120$ different jumble words for the intended word "their", but $\frac{5!}{1!1!1!2!} = 60$ jumble words for the intended word "there". For each jumble word of "there" there are two jumble words for

[5]The term "ordered group" is important. It is not same as the number of ways one can form k groups with the i-th group of size n_i. Say for example for $n = 4$, $k = 2$, $n_1 = n_2 = 2$ with the 4 objects $\{a, b, c, d\}$, $\begin{pmatrix} 4 \\ 2, 2 \end{pmatrix} = \begin{pmatrix} 4 \\ 2 \end{pmatrix}$ =6. This says that there are 6 ways to form 2 ordered groups of size 2 each *viz.* $(\{a, b\}, \{c, d\})$, $(\{a, c\}, \{b, d\})$, $(\{a, d\}, \{b, c\})$, $(\{b, c\}, \{a, d\})$, $(\{b, d\}, \{a, c\})$ and $(\{c, d\}, \{a, c\})$. But the number of possible ways in which one can divide the 4 objects into 2 groups of 2 each is only 3 which are $\{\{a, b\}, \{c, d\}\}$, $\{\{a, c\}, \{b, d\}\}$ and $\{\{a, d\}, \{b, c\}\}$. Similarly say with $n = 7$, $k = 2$, $n_1 = 2$, $n_2 = 2$ and $n_3 = 3$ there are $\begin{pmatrix} 7 \\ 2, 2, 3 \end{pmatrix} = \frac{7!}{2!2!3!}$ =210 many ways of forming 3 ordered groups with respective sizes of 2, 2 and 3, but the number of ways one can divide 7 objects in 3 groups such that 2 groups are of size 2 each and the third one is of size 3 is 210/2=105. The order of the objects within a group does not matter, but the order in which the groups are being formed are counted as distinct even if the contents of the k groups are same.

"their" with "i" in place of one of the two "e"s. $\bigtriangledown$

Example 2.13: Suppose an elevator starts with 9 people who can potentially disembark at 12 different floors above. What is the probability that only one person each disembarking in 3 of the floors and in each of the another 3 floors 2 persons disembarking? First the number of possible ways 9 people can disembark in 12 floors equals 12^9. Now for the given pattern of disembarkment to occur, first the 9 passengers have to be divided in 6 groups with 3 of these groups containing 1 person and the remaining 3 containing 2 persons. This according to the multinomial formula can be done in $\frac{9!}{1!^3 2!^3}$ ways. Now however we have to consider the possible configurations of the floors where the given pattern of disembarkment may take place. For each floor the number of persons disembarking there is either 0, 1 or 2. Also the number of floors where 0 persons disembark equals 6, the number of floors where 1 person disembarks equals 3 and the number of floors where 2 persons disembark is 3, giving the total count of 12 floors. Thus the number of possible floor configurations is same as dividing the 12 floors in 3 groups of 3, 3, and 6 elements, which again according to the multinomial formula is given by $\frac{12!}{3!3!6!}$. Thus the required probability is $\frac{9!}{1!^3 2!^3}\frac{12!}{3!3!6!}12^{-9} = 0.1625$ $\bigtriangledown$

Example 2.14: What is the probability that given 30 people, there are 6 months containing the birthdays of 2 people each, and the other 6 each containing the birthdays of 3 people? Obviously the total number of possible ways in which the birthdays of 30 people can fall in 12 different months equal 12^{30}. For figuring out the number of outcomes favorable to the event of interest, first note that there are $\begin{pmatrix} 12 \\ 6 \end{pmatrix}$ different ways of dividing the 12 months in two groups of 6 each, so that the members of the first group contain birthdays of 2 persons and the members of the first group contain birthdays of 3 persons. Now we shall group the 30 people in two different groups - the first group containing 12 people, so that they can be further divided into 6 groups of 2 each to be assigned to the 6 months chosen to contain the birthdays of 2 people; and the second group containing 18 people, so that they can then be divided into 6 groups of 3 each to be assigned to the 6 months chosen to contain the birthdays of 3 people. The initial two groupings of 30 into 12 and 18 can be done in $\begin{pmatrix} 30 \\ 12 \end{pmatrix}$ ways. Now the 12 can be divided into 6 groups of 2 each in $\frac{12!}{2!^6}$ different ways and the 18 can be divided into 6 groups of 3 each in $\frac{18!}{3!^6}$ different ways. Thus the number of outcomes favorable to the event is given by $\begin{pmatrix} 12 \\ 6 \end{pmatrix}\begin{pmatrix} 30 \\ 12 \end{pmatrix}\frac{12!}{2!^6}\frac{18!}{3!^6} = \frac{12!30!}{2^6 6^6 720^2}$ and the required probability equals $\frac{12!30!}{2^6 6^6 720^2}12^{-30}$. $\bigtriangledown$

Example 2.15: A library has 2 identical copies of Kai Lai Chung's "Elementary Probability Theory with Stochastic Processes" (KLC), 3 identical copies of Hoel, Port and Stone's "Introduction to Probability Theory" (HPS), and 4 identical copies of Feller's Volume I of "An Introduction to Probability Theory and its Applications" (FVI). A monkey is hired to arrange these 9 books on a shelf. What is the probability that one will find the 2 KLC's side by side, 3 HPS's side by side and the 4 FVI's side by side (assuming that the monkey has at least arranged the books one by one on the shelf it was asked to)? The total number of possible ways the 9 books may be arranged side by side in the shelf is given by $\frac{9!}{2!3!4!} = 1260$. The number of ways the event of interest can happen is same as the number of ways the

three blocks of books can be arranged between themselves, which can be done in $3! = 6$ ways. Thus the required probability equals $6/1260 = 0.0048$ $\bigtriangledown$

<u>Formula 5.</u> We have briefly touched upon the issue of indistinguishability of objects in the context of permutation during our discussion of multinomial coefficients in ***Formula 4***. Here we summarize the counting methods involving such indistinguishable objects. To begin with, in the spirit of ***Formula 1***, suppose we are to place k indistinguishable balls in n cells. How many ways can one do that? Let us represent an empty cell by $||$ and a cell containing r balls by putting r $\circ$'s within two bars as $|\underbrace{\circ\cdots\circ}_{r-\text{many}}|$. That is a cell containing one ball is represented by $|\circ|$, a cell containing two balls is represented by $|\circ\circ|$ etc.. Thus a distribution of k indistinguishable balls in n cells may be represented by a sequence of $|$'s and $\circ$'s such as $|\circ|||\circ\circ|\cdots|\circ||\circ\circ\circ||\circ||||$, such that the sequence must a) start and end with a $|$, b) contain $(n+1)$ $|$'s for the n cells, and c) contain k $\circ$'s for the k indistinguishable balls. Hence the number of possible ways of distributing k indistinguishable balls into n cells is same as the number of such sequences. Since the sequence totally must have $n+1+k-2 = n+k-1$ symbols freely choosing their positions within the two $|$'s (and hence that -2) with k of them being a $\circ$ and the remaining $(n-1)$ being a $|$, the possible number of such sequences simply equals the number of ways one can choose $(n-1)$ (k) positions from a possible $(n+k-1)$ and place a $|$ $(\circ)$ in there, and place a $\circ$ $(|)$ in the remaining k $((n-1))$ positions. This can be done in $\begin{pmatrix} n+k-1 \\ n-1 \end{pmatrix}$ $(\equiv \begin{pmatrix} n+k-1 \\ k \end{pmatrix})$ ways, yielding the number of possible ways to distribute k indistinguishable balls in n cells.

The formula $\begin{pmatrix} n+k-1 \\ k \end{pmatrix}$ also applies to the count of number of combinations of k objects chosen from a set of n (distinguishable) objects drawn <u>with replacement</u>. By combination we mean the number of possible groups of k objects, disregarding the order in which the objects were drawn. To see this, again apply the $|\circ||\cdots|||\circ\circ|$ representation with the following interpretation. Represent the n objects with $(n+1)$ $|$'s as $\underbrace{|||\cdots||}_{(n+1)-\text{many}}$, so that for $i = 1, 2, \ldots, n$ the i-th object is represented by the space between the i-th and $(i+1)$-st $|$. Now a combination of k objects drawn with replacement from these n, may be represented by throwing k $\circ$'s within the $(n+1)$ $|$'s as $|\circ|||\circ\circ||\cdots|\circ\circ||$, with the understanding that the number of $\circ$'s within the i-th and $(i+1)$-st $|$ represents the number of times the i-th object has been repeated in the group for $i = 1, 2, \ldots, k$. Thus the number of such possible combinations is same as the number of such sequences that follow the same three constraints a), b) and c) as in the preceding paragraph, which as has been shown there equals $\begin{pmatrix} n+k-1 \\ k \end{pmatrix}$. $\bigtriangledown$

<u>Example 2.16:</u> Let us reconsider the problem in **Example 2.5**. Now instead of 4 rolls of a fair dice, let us slightly change the problem to rolling 4 die simultaneously, and we are still interested in the event, "at least one ace". If the 4 die were distinguishable, say for example of different colors, then this problem is identical to the one discussed in **Example 5** (probabilistically rolling the same dice 4 times is equivalent to one roll of 4 distinguishable

die), and the answer would have been $1-(5/6)^4 = 0.5177$. But what if the 4 die were indistinguishable, say of same color and no other marks to distinguish one from the other? Now the total number of possible outcomes is no longer 6^4. This number now equals the number of ways one can distribute 4 indistinguishable balls in 6 cells. Thus following the foregoing discussion we can compute the total number of possible outcomes as $\binom{6+4-1}{4}$. Similarly the number of ways the complementary event, "no ace" of the event of interest, "at least one ace" can happen is same as distributing 4 indistinguishable balls into 5 cells, which can happen in $\binom{5+4-1}{4}$ ways. Thus by the complementation law (*vide.* §5) the required probability of interest equals $1-\binom{8}{4}\Big/\binom{9}{4} = 0.\dot{4}$. $\triangledown$

Example 2.17: Consider the experiment of rolling $k \geq 6$ indistinguishable die. Suppose we are interested in the probability of the event that none of the faces 1 through 6 are missing in this roll. This event of interest is a special case of distributing k indistinguishable balls in n cells, such that none of the cells are empty, with $n = 6$. For counting the number of ways this can happen let us go back to the $|\circ\circ|\circ|\cdots|\circ\circ\circ|\circ|$ representation of distributing k indistinguishable $\circ$ balls into n $||$ cells. For such a sequence to be a valid representation they must satisfy the three constraints a), b) and c) mentioned in ***Formula 5***. Now for the event of interest to happen the sequence must also satisfy the additional restriction that no two $|$'s must appear side by side, for it represents an empty cell. For this to happen the $(n-1)$ inside $|$'s (recall that we need $(n+1)$ $|$'s to represent n cells, two of which are fixed at either end, leaving the positions of the inside $(n-1)$ $|$'s to be chosen at will) can only appear in the spaces left between two $\circ$'s. Since there are k $\circ$'s there are $(k-1)$ spaces between them, and the $(n-1)$ inside $|$'s can appear only in these positions for honoring the condition "no empty cell", which can be done in $\binom{k-1}{n-1}$ different ways. Thus coming back to the die problem, the number of outcomes favorable to the event, "each face shows up at least once in a roll of k indistinguishable die" equals $\binom{k-1}{5}\Big/\binom{6+k-1}{5} = \prod_{i=1}^{5}\frac{k-i}{k+i}$. $\triangledown$

Example 2.18: Suppose 5 diners enter a restaurant where the chef prepares an item fresh from scratch after an order is placed. The chef that day has provided a menu of 12 items from where the diners can choose their dinners. What is the probability that the chef has to prepare 3 different items for that party of 5? Assume that even if there is more than one request for the same item from a given set of orders, like the one from our party of 5, the chef needs to prepare that item only once. The total number of ways the order for the party of 5 can be placed is same as choosing 5 items out of a total possible 12 with replacement (two or more people can order the same item). This can be done in $\binom{12+5-1}{5}$ ways. (Note that the number of ways the 5 diners can have their choice of items is 12^5. This is the number of arrangements of the 5 selected items, where we are also keeping track of which diner has ordered what item. But as far as the chef is concerned, what matters is only the collective order of 5. If A wanted P, B wanted Q, C wanted R, D wanted R and E wanted P, for the chef it is same as if A wanted Q, B wanted R, C wanted Q, D wanted P and E

wanted Q or any other repeated permutation of {P,Q,R} containing each of these elements at least once. Thus the number of possible collective orders, which is what matters to the chef, is the number of possible groups of 5 one can construct from the menu of 12 items, where repetition is allowed.) Now the event of interest, "the chef has to prepare 3 different items for that party of 5" can happen if and only if the collective order contains 3 distinct items and either one of these 3 items repeated thrice or two of these items repeated twice. 3 distinct items from a menu of 12 can be chosen in $\binom{12}{3}$ ways. Now once 3 distinct items are chosen, two of them can be chosen (to be repeated twice - once in the original distinct 3 and once now) in $\binom{3}{2} = 3$ ways, and one of them can be chosen (to be repeated thrice - once in the original distinct 3 and now twice) in $\binom{3}{1} = 3$ ways. Thus for each $\binom{12}{3}$ ways of choosing 3 distinct items from a menu of 12, there are 3+3=6 ways of generating a collective order of 5, containing each of the first 3 at least once and no other items. Therefore the number of outcomes favorable to the event of interest equals $6\binom{12}{3}$ and the required probability equals $55/182 = 0.3022$. $\bigtriangledown$

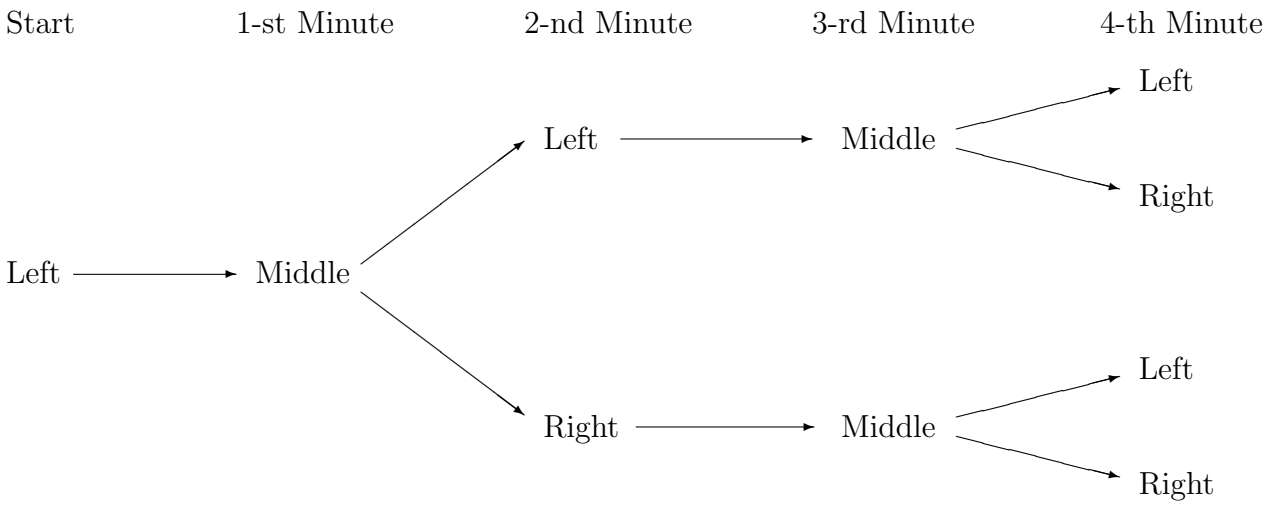
To summarize the counting methods discussed in ***Formulæ* 1** to **5**, first note that the number of possible permutations *i.e.* number of different arrangements, that one can make by drawing k objects <u>with replacement</u> from n (distinguishable) objects is our first combinatorial formula *viz.* n^k. Thus the number of possible permutations and combinations of k objects drawn with and without replacements from a set of n (distinguishable) objects can be summarized in the following table:

No. of Possible	Drawn Without Replacement	Drawn With Replacement
Permutations	$(n)_k = \frac{n!}{(n-k)!}$	n^k
Combinations	$\binom{n}{k} = \frac{n!}{k!(n-k)!}$	$\binom{n+k-1}{k}$

An alternative interpretation of n^k and $\binom{n+k-1}{k}$ are the respective number of ways one can distribute k distinguishable and indistinguishable balls in n cells. Furthermore we are also armed with a permutation formula for the case where some objects are indistinguishable. For $i = 1, 2, \ldots, k$ if there are n_i indistinguishable objects of the i-th kind, where the kinds can be distinguished between themselves, the number of possible ways one can arrange all the $n = \sum_{i=1}^{k} n_i$ objects between themselves is given by $\binom{n}{n_1, \ldots, n_k} = n! \Big/ \prod_{i=1}^{k} n_i!$. Now with the help of these formulæ, and more importantly the reasoning process behind them, one should be able to solve almost any combinatorial probability problem. However we shall close this section only after providing some more examples demonstrating the use of these formulæ and more importantly the nature of combinatorial reasoning.

<u>Example 2.19:</u> A driver driving in a 3-lane one-way road, starting at the left most lane, randomly switches to an adjacent lane every minute. The probability that he is back in the

original left most lane he started with after the 4-th minute is 1/2. This probability can be calculated by a complete enumeration with the help of a tree digram, without getting into attempting to apply any set formula. Thus consider the following tree diagram depicting his lane position after every i-th minute for i=1,2,3,4.



Hence we see that there are a total of 4 possibilities after the 4-th minute, and he is in the left lane in 2 of them. Thus the required probability is 1/2. $\bigtriangledown$

Example 2.20: There are 12 slots in a row in a parking lot, 4 of which are vacant. The chance that they are all adjacent to each other is $0.0\dot{1}\dot{8}$. The number of ways in which 4 slots can remain vacant among 12 is $\begin{pmatrix} 12 \\ 8 \end{pmatrix} = \frac{12!}{8!4!} = 495$. Now the number of ways the 4 vacant slots can be adjacent to each other is found by direct enumeration, which can happen if and only if the positions of the empty slots are one of the following {1,2,3,4; 2,3,4,5; ... 8,9,10,11; 9,10,11,12}, consisting of 9 cases favorable to the event. Thus the required probability is 9/495=$0.0\dot{1}\dot{8}$. $\bigtriangledown$

Example 2.21: n students are assigned at random to n advisers. The probability that exactly one adviser does not have any student with her is $\frac{n(n-1)n!}{2n^n}$. This is because the total number of possible adviser-student assignment equals n^n. Now if exactly one of the advisers does not have any student with her, there must be exactly one adviser who is advising two students, and the remaining $(n-2)$ advisers are advising exactly one student each. The number of ways one can choose one adviser with no student and another adviser with two students is $(n)_2 = n(n-1)$. The remaining $(n-2)$ advisers must get one student each from a total pool of n students. This can be done in $(n)_{n-2} = n!/2$ ways. Thus the required probability equals $\frac{n(n-1)n!}{2n^n}$. $\bigtriangledown$

Example 2.22: One of the CNC machines in a factory is handled by one of the 4 operators. If not programmed properly the machine halts. The same operator, but not known which one, was in-charge during at least 3 such halts among the last 4. Based on this evidence can it be said that the concerned operator is incompetent? The total number of possible ways the 4 operators could have been in-charge during the 4 halts is 4^4. The number of ways in which a given particular operator could have been in-charge during exactly 3 of

them is $\binom{4}{3}\binom{3}{1}$ ($\binom{4}{3}$ ways of choosing the 3 halts of the 4 for the particular operator and $\binom{3}{1}$ way of choosing the operator who was in-charge during the other halt); and the number of ways in which that operator could have been in-charge during all 4 of the halts $= 1$. Thus given a particular operator, the number of ways he could have been in-charge in at least 3 of 4 such halts equals 13. But since it is not known which operator it was, who was in-charge during the 3 or more halts, that particular operator can further be chosen in 4 ways. Thus the event of interest, "the same operator was in-charge during at least 3 of the last 4 halts" can happen in $4 \times 13 = 52$ different ways, and thus the required probability of interest equals $52/4^4$=0.203125. This is not such a negligible chance after all, and thus branding that particular operator, whosoever it might have been, as incompetent is possibly not very fair. $\bigtriangledown$

Example 2.23: $2k$ shoes are randomly drawn out from a shoe-closet containing n pairs of shoes, and we are interested in the probability of finding at least one original pair among them. We shall take the complementary route and attempt to find the probability of finding not a single one of the original pairs. $2k$ shoes can be drawn from the n pairs or $2n$ shoes in $\binom{2n}{2k}$ ways. Now if there is not a single one of the original pairs among them, all of the $2k$ shoes must have been drawn from a collection of n shoes, consisting of one shoe from each of the n pairs, which can be done in $\binom{n}{2k}$ ways. But now there are exactly two possibilities for each of the $2k$ shoes, which are coming from one of the shoes of the n pairs, say the left or the right of the corresponding pair. This gives rise to $\underbrace{2 \times 2 \times \cdots \times 2}_{2k-\text{times}} = 2^{2k}$ possibilities. Thus the number ways in which the event, "not a single pair" can happen equals $\binom{n}{2k}2^{2k}$,[6] and hence by the complementation law (*vide.* §5) the probability of "at

[6]Typically counts in such combinatorial problems may be obtained using several different arguments, and in order to get the count correct, it may not be a bad idea to argue the same counts in different ways to ensure that we are after all getting the same counts using different arguments. Say in this example, we can alternatively argue the number of favorable cases to the event "not a single pair" as follows. Suppose among the $2k$ shoes there are exactly l which are of left foot and the remaining $2k-l$ are of right foot. So the possible values l can take would run from 0, 1,... to $2k$, and each of these events are mutually exclusive, so that the total number of favorable cases would equal sum of such counts. Now the number ways the l-th one of these events can happen, so that there is no pair is $\binom{n}{l}\binom{n-l}{2k-l}$ (first choose the l left foot shoes from the total possible n, and then choose the $2k-l$ right foot shoes from those pairs for which the corresponding left foot shoe have not already been chosen, of which there are $n-l$ such). Thus the number of cases favorable to the event equals $\sum_{l=0}^{2k}\binom{n}{l}\binom{n-l}{2k-l} = \sum_{l=0}^{2k}\frac{n!}{l!(2k-l)!(n-2k)!} = \sum_{l=0}^{2k}\frac{n!}{(2k)!(n-2k)!}\frac{(2k)!}{l!(2k-l)!}$ $= \sum_{l=0}^{2k}\binom{n}{2k}\binom{2k}{l} = \binom{n}{2k}2^{2k}$, coinciding with the previous argument.

least one of the original pairs" equals $1 - \dfrac{\binom{n}{2k} 2^{2k}}{\binom{2n}{2k}}$. $\triangledown$

Example 2.24: What is the probability that the birthdays of 6 people will fall in exactly 2 different calendar months? The total number of ways in which the birthdays of 6 people can be assigned to the 12 different calendar months equals 12^6. Now if all these 6 birthdays are falling in exactly 2 different calendar months; first the number of such possible pairs of months equals $\binom{12}{2}$; and then the number of ways one can distribute the 6 birthdays in these two chosen months equals $\binom{6}{1}+\binom{6}{2}+\binom{6}{3}+\binom{6}{4}+\binom{6}{5}$ (choose k birthdays out of 6 and assign them to the first month and the remaining $6-k$ to the second month - since each month must contain at least one birthday, the possible values k can assume are 1, 2, 3, 4, and 5) $= \left\{\binom{6}{0}+\binom{6}{1}+\binom{6}{2}+\binom{6}{3}+\binom{6}{4}+\binom{6}{5}+\binom{6}{6}\right\} - 2$ $= 2^6 - 2$ (an alternative way of arguing this $2^6 - 2$ could be as follows - for each of the 6 birthdays there are 2 choices, thus the total number of ways in which the 6 birthdays can be assigned to the 2 selected months equals 2^6, but among them there are 2 cases where all the 6 birthdays are being assigned to a single month, therefore the number of ways one can assign 6 birthdays to the 2 selected months such that each month contains at least one birthday must equal $2^6 - 2$). Thus the number of cases favorable to the event equals $\binom{12}{2}(2^6-2)$ and the required probability is $\binom{12}{2}(2^6-2)\,12^{-6}$. $\triangledown$

Example 2.25: In a population of $n+1$ individuals, a person, called the progenitor, sends out an e-mail at random to k different individuals, each of whom in turn again forwards the e-mail at random to k other individuals and so on. That is at every step, each of the recipients of the e-mail forwards it to k of the n other individuals at random. We are interested in finding the probability of the e-mail not relayed back to the progenitor even after r steps of circulation. The number of possible recipients from the progenitor is $\binom{n}{k}$. The number of possible choices each one of these k recipients has after the first step of circulation is again $\binom{n}{k}$, and thus the number of possible ways this first stage recipients can forward the e-mail equals $\underbrace{\binom{n}{k} \times \cdots \times \binom{n}{k}}_{k-\text{times}} = \binom{n}{k}^k$. Therefore after the second step of circulation the total number of possible configurations equals $\binom{n}{k}^{1+k}$. Now there are $k \times k = k^2$ many second-stage recipients each one of whom can forward the e-mail to

$\begin{pmatrix} n \\ k \end{pmatrix}$ possible recipients yielding a possible $\begin{pmatrix} n \\ k \end{pmatrix}^{k^2}$ many third-stage recipients after 3 steps of circulations and $\begin{pmatrix} n \\ k \end{pmatrix}^{1+k+k^2}$ many total possible configurations. Proceeding in this manner one can see that after the e-mail has been circulated through $r-1$ steps, at the r-th step of circulation the number of senders equal k^{r-1} who can collectively make $\begin{pmatrix} n \\ k \end{pmatrix}^{k^{r-1}}$ many choices. Thus the total number of possible configurations after the e-mail has been circulated through r-steps equals $\begin{pmatrix} n \\ k \end{pmatrix}^{1+k+k^2+\cdots+k^{r-1}} = \begin{pmatrix} n \\ k \end{pmatrix}^{\frac{k^r-1}{k-1}}$. Now the e-mail does not come back to the progenitor in any of these r steps of circulation if and only if none of, starting from the k recipients of the progenitor after the first step of circulation to the k^{r-1} recipients after $r-1$ steps of circulation, sends it to the progenitor, or in other words each of these recipients/senders at every step makes a choice of forwarding the e-mail to k individuals from a total of $n-1$ instead of the original n. Thus the number of ways the e-mail can get forwarded through the second, third, ..., r-th step avoiding the progenitor equals $\begin{pmatrix} n-1 \\ k \end{pmatrix}^{k+k^2+\cdots+k^{r-1}} = \begin{pmatrix} n-1 \\ k \end{pmatrix}^{\frac{k^r-1}{k-1}-1} = \begin{pmatrix} n-1 \\ k \end{pmatrix}^{\frac{k^r-k}{k-1}}$. The number of choices for the progenitor remains the same, namely $\begin{pmatrix} n \\ k \end{pmatrix}$. Thus the number of possible outcomes favorable to the event of interest equals $\begin{pmatrix} n \\ k \end{pmatrix}\begin{pmatrix} n-1 \\ k \end{pmatrix}^{\frac{k^r-k}{k-1}}$, yielding the probability of interest as $\left\{\begin{pmatrix} n-1 \\ k \end{pmatrix} \Big/ \begin{pmatrix} n \\ k \end{pmatrix}\right\}^{\frac{k^r-k}{k-1}} = \left\{\frac{(n-1)!}{k!(n-k-1)!}\frac{k!(n-k)!}{n!}\right\}^{\frac{k^r-k}{k-1}} = \left(\frac{n-k}{n}\right)^{\frac{k^r-k}{k-1}} = \left(1-\frac{k}{n}\right)^{\frac{k^r-k}{k-1}}$. $\bigtriangledown$

Example 2.26: n two member teams, consisting of a junior and a senior member, are broken down and then again regrouped at random to form n two member teams. We are interested in finding the probability that each of this regrouped n two member teams again contains a junior and a senior member each. The first problem is to find the number of possible n two member teams that one can form from these $2n$ individuals. The number of possible ordered groups of 2 that can be formed is given by $\begin{pmatrix} 2n \\ \underbrace{2,\ldots,2}_{n-\text{times}} \end{pmatrix} = (2n)!/2^n$. A possible such grouping gives n two member teams alright, but $(2n)!/2^n$ contains all such ordered groupings. That is even if the n teams were same, if they were constructed following a different order they will be counted as distinct in the counts of $(2n)!/2^n$, while we are only interested in the possible number of ways to form n groups each containing two members, and not in the order in which these groups are formed. This situation is analogous to our interest in combination, while a straight-forward reasoning towards that end takes us first to the number of permutations. Hence this problem is also resolved exactly in the similar manner. Given a configuration of n groups each containing 2 members, how many times is this configuration counted in that count of $(2n)!/2^n$? It is same as the number of possible

ways one can arrange these n teams among themselves with each arrangement leading to a different order of formation, which are counted as distinct in the count of $(2n)!/2^n$. Now the number of ways one can arrange the n teams among themselves equals $n!$ and therefore the number of possible n two member teams that one can form with $2n$ individuals may be obtained by dividing the number of possible ordered groups ($= (2n)!/2^n$) by the number of possible orders for the same configuration of n two member teams, which equals $n!$. Hence the total number of possible outcomes is given by $\frac{(2n)!}{n!2^n}$.[7] For the number of possible outcomes favorable to the event of interest, "each of the regrouped n two member teams contains a junior and a senior member each", assign and fix position numbers 1, 2, ..., n to the n senior members in any order you please. Now the number of possible teams that can be formed with the senior members and one of the junior members, is same as the number of ways one can arrange the n junior members in the positions 1, 2, ..., n assigned to the n senior members, which can be done in $n!$ ways. Thus the required probability of interest equals $\frac{n!^2 2^n}{(2n)!}$. $\bigtriangledown$

Example 2.27: A sample of size n is drawn with replacement from a population containing N individuals. We are interested in computing the probability that among the chosen n exactly m individuals are distinct. Note that the exact order in which the individuals appear in the sample is immaterial and we are only interested in the so-called unordered sample. First note that the number of such possible (unordered) samples equals the number of possible groups of size n one can form by choosing from N individuals with replacement, which as argued in ***Formula 5*** equals $\begin{pmatrix} N+n-1 \\ n \end{pmatrix}$. The number of ways one can choose the m distinct individuals to appear in the sample equals $\begin{pmatrix} N \\ m \end{pmatrix}$. Now the sample must be such that these are the only individuals appearing in the sample at least once and the other $N-m$ are not. Coming back to the $||\circ\circ|\cdots||\circ|$ representation, this means that once the m positions among the N available spaces between two consecutive $|$'s (representing the N individuals in the population) have been chosen, which can be done in $\begin{pmatrix} N \\ m \end{pmatrix}$ ways; all the k $\circ$'s representing the k draws must be distributed within these m spaces such that none of these m spaces are empty, ensuring that all these m have appeared at least once and none of the remaining $N-m$ appearing even once. The last clause (appearing after the semi-colon) can be accomplished in $\begin{pmatrix} k-1 \\ m-1 \end{pmatrix}$ ways, because there are $(k-1)$ spaces between the k $\circ$'s enclosed between two $|$'s at the either end, and now $(m-1)$ $|$'s are to be placed in these $(k-1)$ spaces between two consecutive $\circ$'s ensuring that none of these m inter $|$-spaces are

[7] An alternative way of arguing this number is as follows. Arrange the $2n$ individuals in a row and then form n two member teams by pairing up the individuals in the first and second positions, third and fourth positions etc. $(2n-1)$-st and $2n$-th positions. Now the number of ways $2n$ individuals can be arranged in a row is given by $(2n)!$. But now among them the adjacent groups of two used to form the n teams can be arranged between themselves in $n!$ ways, and further the positions of the two individuals in the same team can be swapped in 2 ways, which for n teams give a total of 2^n possibilities. That is if one considers any of the $(2n)!$ arrangements, corresponding to it, there are $n!2^n$ possible arrangements which yield the same n two member teams but which are counted as distinct in the $(2n)!$ possible arrangements. Hence the number of possible n two member teams must equal $\frac{(2n)!}{n!2^n}$.

empty. (Recall that in **Example 17** we have already dealt with this issue of distributing k indistinguishable balls into n cells such that none of the cells are empty, for which the answer was $\binom{k-1}{n-1}$. Here also the problem is identical. We are to distribute k (indistinguishable) balls into m cells such that none of them are empty, which as before can be done in $\binom{k-1}{m-1}$ ways.) Hence the number of outcomes favorable to the event equals $\binom{N}{m}\binom{k-1}{m-1}$ and the required probability of interest is $\left\{\binom{N}{m}\binom{k-1}{m-1}\right\} \Big/ \binom{N+n-1}{n}$. $\bigtriangledown$

Example 2.28: One way of testing for randomness in a given sequence of symbols is accomplished by considering the number of *runs*. A run is an unbroken sequence of like symbols. Suppose the sequence consists of two symbols α and β. Then a typical sequence looks like $\alpha\alpha\beta\alpha\beta\beta\beta\alpha\alpha$, which contains 5 runs. The first run consists of two α's, second run consists of one β, third run consists of one α, fourth run consists of two β's and the fifth run consists of two α's, and thus the sequence contains 5 runs in total. If there are too many runs in a sequence that shows an alternating pattern, while if there are too few runs that shows a clustering pattern. Thus one can investigate the issue of whether the symbols appearing in a sequence are random or not by studying the behavior of the number of runs in them. Here we shall confine ourselves to two-symbol sequences, say α and β.

Suppose we have a sequence of length n consisting of n_1 α's and n_2 β's. Then the minimum number of runs that the sequence must contain is 2 (all n_1 α's together and all the n_2 β's together) and the maximum is $2n_1$ if $n_1 = n_2$ and $2 \times \text{Minimum}\{n_1, n_2\} + 1$, otherwise. If $n_1 = n_2$ the number of runs will be maximum if the α's and β's appear alternatingly giving rise to $2n_1$ runs. For the case $n_1 \neq n_2$, without loss of generality suppose $n_1 < n_2$. Then the number of runs will be maximum if there is at least one β within each of the two consecutive α's. There are $n_1 - 1$ spaces between the n_1 α's and we have enough β's to place at least one each in each of these $n_1 - 1$ spaces, leaving at least two more β's with at least one placed before the first α and at least one placed after the last α yielding a maximum number of runs of $2n_1 + 1$.

Now suppose we have r_1 α-runs and r_2 β-runs, yielding a total of $r = r_1 + r_2$ runs. Note that if there are r_1 α-runs there are $r_1 - 1$ spaces between the r_1 α-runs which must be filled with the β-runs. There might also be a β-run before the first α-run and/or after the last α-run. Thus if there are r_1 α-runs, r_2, the number of β-runs must equal either r_1 or $r_1 \pm 1$, and vice-versa. Thus for considering the distribution of the total number of runs we have to deal with the two cases separately *viz.* r is even and odd.

First suppose $r = 2k$ an even number. This can happen if and only if the number of α-runs = the number of β-runs = k. The total number of ways n_1 α's and n_2 β's can appear in a sequence of length n is same as the number of ways one can choose the n_1 positions (n_2 positions) out of the total possible n for the n_1 α's (n_2 β's), which can be done in $\binom{n}{n_1}$

$(\equiv \begin{pmatrix} n \\ n_2 \end{pmatrix})$ ways. Now the number of ways one can distribute the n_1 α's into its k runs is same as the number of ways one can distribute n_1 indistinguishable balls (since the n_1 α's are indistinguishable) into k cells such that none of the cells are empty, which according to **Example 17** can be done in $\begin{pmatrix} n_1 - 1 \\ k - 1 \end{pmatrix}$ ways. Similarly the number of ways one can distribute n_2 β's into k runs is same as $\begin{pmatrix} n_2 - 1 \\ k - 1 \end{pmatrix}$, and each way of distributing the n_1 α's into k runs is associated with each way of distributing the n_2 β's into k runs. Furthermore if the number of runs is even, the sequence must either start with an α-run and end with a β-run or start with a β-run and end with an α-run, and for each of these configurations there are $\begin{pmatrix} n_1 - 1 \\ k - 1 \end{pmatrix}\begin{pmatrix} n_2 - 1 \\ k - 1 \end{pmatrix}$ ways of distributing n_1 α's and n_2 β's into k runs each. Therefore the number of possible ways the total number of runs can equal $2k$ is $2\begin{pmatrix} n_1 - 1 \\ k - 1 \end{pmatrix}\begin{pmatrix} n_2 - 1 \\ k - 1 \end{pmatrix}$, and hence the required probability of interest is $\left\{2\begin{pmatrix} n_1 - 1 \\ k - 1 \end{pmatrix}\begin{pmatrix} n_2 - 1 \\ k - 1 \end{pmatrix}\right\} \Big/ \begin{pmatrix} n \\ n_1 \end{pmatrix}$.

Now suppose $r = 2k + 1$. r can take the value $2k + 1$ if and only if either $r_1 = k$ & $r_2 = k + 1$ or $r_1 = k + 1$ & $r_2 = k$. This break-up is analogous to the sequence starting with an α-run or a β-run as in the previous (even) case. Following arguments similar to above $r_1 = k$ & $r_2 = k + 1$ can happen in $\begin{pmatrix} n_1 - 1 \\ k - 1 \end{pmatrix}\begin{pmatrix} n_2 - 1 \\ k \end{pmatrix}$ ways, and $r_1 = k + 1$ & $r_2 = k$ can happen in $\begin{pmatrix} n_1 - 1 \\ k \end{pmatrix}\begin{pmatrix} n_2 - 1 \\ k - 1 \end{pmatrix}$ ways. Thus the required probability of interest is $\left\{\begin{pmatrix} n_1 - 1 \\ k - 1 \end{pmatrix}\begin{pmatrix} n_2 - 1 \\ k \end{pmatrix} + \begin{pmatrix} n_1 - 1 \\ k \end{pmatrix}\begin{pmatrix} n_2 - 1 \\ k - 1 \end{pmatrix}\right\} \Big/ \begin{pmatrix} n \\ n_1 \end{pmatrix}$. ∇

2.5 Probability Laws

In this section we take up the cue left after the formal mathematical definition of Probability given in ***Definition 3*** in §3. §4 showed how logically probabilities may be assigned to non-trivial events ($A \in \mathcal{A} \neq \phi$ or Ω) for a finite Ω with all elementary outcomes being *equally likely*. As is obvious, such an assumption severely limits the scope of application of Probability theory. Thus in this section we explore the mathematical consequences the $P(\cdot)$ of ***Definition 3*** must face in general, which are termed as Probability Laws. Apart from their importance in the mathematical theory of Probability, from the application point of view, these laws are also very useful in evaluating probabilities of events in situations where they must be argued out using probabilistic reasoning and numerical probability values of some other more elementary events. A very mild flavor of this approach towards probability calculation can already be found in a couple of Examples worked out in §4 with due reference given to this section, though care was taken in not heavily using these laws without introducing them first, as will be done with the examples in this section.

There are three basic laws that the probability function $P(\cdot)$ of ***Definition 2.3*** must abide by. These are called ***complementation law***, ***addition law*** and ***multiplication law***. Apart from these these three laws, $P(\cdot)$ also has two important properties called the ***monotonicity property*** and ***continuity property*** which are useful for proving theoretical results. Of these five, multiplication law requires the notion of a new concept called **conditional probability** and will thus be taken up in a separate subsection later in this section.

Complementation Law: $P(A^c) = 1 - P(A)$.
Proof:
$P(A^c)$

$= P(A \cup A^c) - P(A)$ (since $A \cap A^c = \phi$, by **iii'** of ***Definition 3***, $P(A \cup A^c) = P(A) + P(A^c)$)

$= P(\Omega) - P(A)$ (by the definition of A^c)

$= 1 - P(A)$ (by **i** of ***Definition 3***) $\bigtriangledown$

For applications of the complementation law for computing probabilities, see **Examples 5, 8, 16** and **23** of §4.

Addition Law: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.
Proof:
$P(A \cup B)$

$= P(\{A \cap B^c\} \cup \{A \cap B\} \cup \{A^c \cap B\})$ (since $A \cup B$ is a union of these three components)

$= P(A \cap B^c) + P(A \cap B) + P(A^c \cap B)$ (by **iii'** of ***Definition 3***, as these three sets are disjoint)

$= \{P(A \cap B^c) + P(A \cap B)\} + \{P(A^c \cap B) + P(A \cap B)\} - P(A \cap B)$

$= P(A) + P(B) - P(A \cap B)$ (by **iii'** of ***Definition 3***, as $A = \{A \cap B^c\} \cup \{A \cap B\}$, and $B = \{A^c \cap B\} \cup \{A \cap B\}$ are mutually exclusive disjointifications of A and B respectively) $\bigtriangledown$

Example 2.29: Suppose in a batch of 50 MBA students, 30 are taking either Strategic Management or Services Management, 10 are taking both and 15 are taking Strategic Management. We are interested in calculating the probability of a randomly selected student taking Services Management. For the randomly selected student, if A and B respectively denote the events "taking Strategic Management" and "taking Services Management", then it is given that $P(A \cup B) = 0.6$, $P(A \cap B) = 0.2$ and $P(A) = 0.3$, and we are to find $P(B)$. A straight forward application of the addition law yields $P(B) = P(A \cup B)$ - $P(A)$ + $P(A \cap B)$ = 0.6 - 0.3 + 0.2 = 0.5. It would be instructive to note that the number of students taking only Services Management and not Strategic Management is 30-15=15, and adding 10 to that (who are taking both) yields that there are 25 students taking Services Management, and thus the required probability is again found to be 0.5 by this direct method. However as is evident, it is much easier to arrive at the answer by mechanically applying the addition law. For more complex problems direct reasoning many times proves to be difficult, which are more easily tackled by applying the formulæ of probability laws. $\bigtriangledown$

The addition law can be easily generalized for unions of n events $A_1 \cup \cdots \cup A_n$ as follows. Let $S_1 = \sum_{i_1} p_{i_1}$, $S_2 = \sum_{i_1 \neq i_2} p_{i_1 i_2}$, ..., $S_k = \sum_{i_1 \neq \cdots \neq i_k} p_{i_1 \ldots i_k}$, ... $S_n = \sum_{i_1 \neq \cdots \neq i_n} p_{i_1 \ldots i_n}$, where $p_{i_1 \ldots i_k} = P(A_{i_1} \cap \cdots \cap A_{i_k})$ for $k = 1, \ldots, n$. Then

$$P(A_1 \cup \cdots \cup A_n) = S_1 - S_2 + S_3 - \cdots + (-1)^{n+1} S_n = \sum_{k=1}^{n} (-1)^{k+1} S_k \tag{2}$$

Equation (2) can be proved by induction on n and the addition law, but a direct proof of this is a little more illuminating. Consider a sample point $\omega \in \cup_{i=1}^{n} A_i$, which belongs to exactly $1 \leq r \leq n$ of the A_i's. Without loss of generality suppose the r sets that ω belongs to are $A_1, \ldots, A_r$ so that it does not belong to $A_{r+1}, \ldots, A_n$. Now $P(\{\omega\}) = p$ (say) contributes exactly once in the l.h.s. of (2), while the number of times its contribution is counted in the r.h.s. requires some calculation. If we can show that this number also exactly equals 1, then that will establish the validity of (2). p contributes - r times in S_1, since ω belongs to r of the A_i's; $\binom{r}{2}$ times in S_2; and in general it contributes $\binom{r}{k}$ times in S_k for $1 \leq k \leq r$ and 0 times in S_k for $r+1 \leq k \leq n$. Thus the total number of times p contributes in the r.h.s. of (2) equals

$$\binom{r}{1} - \binom{r}{2} + \cdots + (-1)^{r+1} \binom{r}{r} = \sum_{k=1}^{r} (-1)^{k+1} \binom{r}{k} = 1 - \sum_{k=0}^{r} (-1)^k \binom{r}{k} = 1 - (1-1)^r = 1$$

Example 2.30: Suppose after the graduation ceremony, n military cadets throw their hats in the air and then each one randomly picks up a hat upon their return to the ground. We are interested in the probability that there will be at least one match, in the sense of a cadet getting his/her own hat back. Let A_i denote the event, "i-th cadet got his/her own hat back". Then the event of interest is given by $\cup_{i=1}^{n} A_i$ whose probability can now be determined using (2). In order to apply (2) we need to figure out $p_{i_1 \ldots i_k}$, for a given $i_1 \neq \cdots \neq i_k$ for $k = 1, \ldots, n$. $p_{i_1 \ldots i_k}$ is the probability of the event, "i_1-th, i_2-th, ..., i_k-th cadet got his/her own hat back", which is computed as follows. The total number of ways the n hats can be picked up by the n cadets is given by $n!$, while out of these the number of cases where the i_1-th, i_2-th, ..., i_k-th cadet picks up his/her own hat is given by $(n-k)!$, yielding $p_{i_1 \ldots i_k} = (n-k)!/n!$. Note that $p_{i_1 \ldots i_k}$ does not depend on the exact sequence $i_1, \ldots, i_k$, and thus $S_k = \binom{n}{k} \frac{(n-k)!}{n!}$ (since S_k has $\binom{n}{k}$ many terms in the summation) $= 1/k!$. Therefore the probability of the event of interest, "at least one match" is given by $1 - \frac{1}{2!} + \frac{1}{3!} - \cdots + (-1)^{n+1} \frac{1}{n!} = 1 - \left(1 - 1 + \frac{1}{2!} - \frac{1}{3!} + \cdots + (-1)^n \frac{1}{n!}\right) \approx 1 - e^{-1} \approx 0.63212$. Actually one gets to this magic number 0.63212 of matching probability pretty fast for n as small as 8, which shows that the probability of at least one match or the complementary event, "no match" is practically independent of n, which is quite surprising! $\triangledown$

Equation (2) requires knowledge of probabilities of intersections for calculating probabilities of unions. The next law, called the multiplication law helps us compute the probabilities of intersections. However as mentioned in the beginning of §5, this requires introduction of an additional concept called conditional probability. Before that however we shall first discuss a couple of more properties of $P(\cdot)$. Unlike the three laws, these properties are not directly

useful in computing probabilities, but they play very important role in probability theory and mathematical statistics and will be required in later chapters. Thus we shall discuss them here though on the surface they might appear rather theoretical in nature without any immediate practical benefit.

Monotonicity Property: If $A \subseteq B$, $P(A) \leq P(B)$.
Proof: Since $A \subseteq B$, $B = A \cup (A^c \cap B)$. Since $A \cap (A^c \cap B) = \phi$, by **iii'** of ***Definition 3***, $P(B) = P(A) + P(A^c \cap B) \geq P(A)$, as $P(A^c \cap B) \geq 0$ by **ii** of ***Definition 3***. $\bigtriangledown$

Continuity Property (i): If $A_1 \subseteq A_2 \subseteq \cdots$ and $A = \cup_{n=1}^{\infty} A_n \stackrel{\text{def.}}{=} \lim_{n\to\infty} A_n$, then $P(A) = P(\lim_{n\to\infty} A_n) = \lim_{n\to\infty} P(A_n)$.
(ii): If $A_1 \supseteq A_2 \supseteq \cdots$ and $A = \cap_{n=1}^{\infty} A_n \stackrel{\text{def.}}{=} \lim_{n\to\infty} A_n$, then $P(A) = P(\lim_{n\to\infty} A_n) = \lim_{n\to\infty} P(A_n)$.
Proof (i): Let $B_1 = A_1$ and for $n \geq 2$ let $B_n = A_n - A_{n-1} = A_n \cap A_{n-1}^c$. Then since $A_1 \subseteq A_2 \subseteq \cdots$, $B_m \cap B_n = \phi$ for $m \neq n$ and $A_n = \cup_{k=1}^{n} B_k$, so that by **iii'** of ***Definition 3***, $P(A_n) = \sum_{k=1}^{n} P(B_k)$. Also $A = \cup_{n=1}^{\infty} A_n = \cup_{n=1}^{\infty} (\cup_{k=1}^{n} B_k) = \cup_{n=1}^{\infty} B_n$. Now

$$\begin{aligned}
P(\lim_{n\to\infty} A_n) &= P(A) \\
&= \sum_{n=1}^{\infty} P(B_n) \text{ (by } \textbf{iii} \text{ of } \textbf{\textit{Definition 3}}\text{, since } A = \cup_{n=1}^{\infty} B_n \text{ and for } m \neq n, B_m \cap B_n = \phi) \\
&= \lim_{n\to\infty} \sum_{k=1}^{n} P(B_k) \text{ (by the definition of infinite series)} \\
&= \lim_{n\to\infty} P(A_n) \text{ (since } P(A_n) = \sum_{k=1}^{n} P(B_k)) \qquad \bigtriangledown
\end{aligned}$$

(ii): For $A_1 \supseteq A_2 \supseteq \cdots$ and $A = \cap_{n=1}^{\infty} A_n$, $A_1^c \subseteq A_2^c \subseteq \cdots$ and $A^c = \cup_{n=1}^{\infty} A_n^c$ by DeMorgan's law. Therefore by continuity property **(i)**, $P(A^c) = \lim_{n\to\infty} P(A_n^c)$, so that $P(\lim_{n\to\infty} A_n) = P(A) = 1 - P(A^c) = \lim_{n\to\infty}(1 - P(A_n^c)) = \lim_{n\to\infty} P(A_n)$. $\bigtriangledown$

Above is called the continuity property for the following reason. A real-valued function of real numbers $f(\cdot)$ is continuous iff for every sequence $x_n \to x$, $f(x_n) \to f(x)$, or in other words the limit and $f(\cdot)$ can be interchanged iff $f(\cdot)$ is continuous. The domain of the probability function $P(\cdot)$ being sets, instead of real numbers, the continuity property ensures that limit and $P(\cdot)$ also can be interchanged provided the sequence of sets has a limit. If the sets are increasing as in **(i)** or decreasing as in **(ii)**, their limits always exist and are naturally defined as their union and intersection respectively. For an arbitrary sequence of sets A_n, their limit is defined as follows. Let $B_n = \cup_{k=n}^{\infty} A_k$ and $C_n = \cap_{k=n}^{\infty} A_k$. Note that B_n is a decreasing sequence of sets as in **(ii)** and they always have limit $B = \cap_{n=1}^{\infty} B_n$; and likewise C_n is an increasing sequence of sets as in **(i)** and they always have limit $C = \cup_{n=1}^{\infty} C_n$. The set $B = \cap_{n=1}^{\infty} \cup_{k=n}^{\infty} A_k$ is called lim sup A_n and $C = \cup_{n=1}^{\infty} \cap_{k=n}^{\infty} A_k$ is called lim inf A_n. The set B consists of those elements which occur in infinitely many of the A_n's while the set C consists of those elements which occur in all but finitely many A_n's. Now the sequence of sets A_n is said to have a limit if these two sets coincide *i.e.* if $B = C$ or lim sup A_n = lim inf A_n. If an arbitrary sequence of sets A_n has a limit, then again it can be shown that Probability of this limiting set is same as the limit of $P(A_n)$, the proof of which easily follows from the above continuity property and being sort of unnecessary for these elementary notes, is left as an exercise for the more mathematically oriented readers.

2.5.1 Conditional Probability

In a way probability of non-trivial sets attempts to systematically quantify our level ignorance about an event. Thus the numerical value of the probability of an event depends on our state of knowledge about a chance experiment. For the same event its appraised probability will in general be different in two instances with different states of knowledge regarding the chance experiment. This notion of letting the probability of an event depend on the state of knowledge is crystallized by introducing the concept of conditional probability. We begin our discussion of conditional probability with a loose informal definition.

Definition 2.4: **Conditional Probability** of an event A given that one knows that B has already occurred is same as the probability of A computed in the restricted sample space B, instead of the original sample space Ω, and is written as $P(A|B)$ (read as, "probability of A given B").

Few simple examples will help illustrate this notion of conditional probability given in the above definition.

Example 2.31: Suppose a student is selected at random in a Statistics class being taken by both the MBA and Ph.D. students. Along with the degree programme a student is in, the gender-wise distribution of the number of students in this Statistics class is as follows:

Degree→ Gender↓	MBA	Ph.D.
Female	20	10
Male	40	10

Let A be the event that the selected student is doing MBA and B be the event that she is a female. Then the unconditional probability of the event A is 60/80=3/4, which is computed using (1) with $N = 80$ and $n = 60$ for the entire sample space of 80 students. But now suppose we have the additional information that the chosen student is a female. In light of this information the chance of the chosen student doing MBA might change, and is calculated using ***Definition 4*** as follows. The formula that is used for calculation of this $P(A|B)$ is still same as that given equation in (1), but now for its n and N, instead of the earlier (unconditional) consideration of the entire sample space of 80 students, our sample space gets reduced to B comprising only the 30 female students. The logic behind this argument is that in presence of the information B, the 50 male students of the class should become irrelevant to the probability calculation and should not figure into our consideration. Thus now with this reduced B as our sample space, its $N = 30$ and within this 30, n, the number of cases favorable to the event A, that the student is doing MBA, is just 20, and thus $P(A|B) = 20/30 = 2/3$. Note that the conditional probability of A gets slightly reduced compared to the unconditional case, because though the proportion of students doing MBA is much larger compared to Ph.D. for both the genders, they are more so in case of the males ($P(A|B^c) = 4/5$) compared to the females, and as a result for the female population with B as the sample space, $P(A|B)$ gets reduced compared to the overall unconditional $P(A)$. Conditional probability helps one put quantitative numbers behind such qualitative analysis.

$\bigtriangledown$

Example 2.32: Consider a population of families with two children. A family is chosen at random from this population and one of the children in this family is found to be a girl. We are interested in finding the probability that the other child in the family is also a girl. The population of such families may be characterized as $\Omega = \{gg, gb, bg, bb\}$, where g stands for a girl and b stands for a boy[8]. Now the given event say B, "one of the children in the chosen family is a girl", equals $\{gg, gb, bg\}$, and given this sample space (instead of the original Ω) we are interested in the probability of the event A, "the other child in the family is also a girl", which is given by $\{gg\}$. This conditional probability $P(A|B) = 1/3$ (and not $1/2$, as some of you might have thought!). $\bigtriangledown$

Example 2.33: Let us again reconsider **Example 2.5**, where we were concerned with the probability of the event A, "at least one ace" in four throws of a fair dice. In **Example 5** it was shown that the unconditional probability of this event equals $1 - (5^4/6^4)$. But now suppose we have the additional information B, "no two throws showed the same face" and are still interested in the probability of event A. As usual the first step is making the counting problem easier by considering the complementary event, "no ace". But now in presence of the information B we must reconsider our sample space and need to redo the calculation of n and N of (1). The sample space B now consists of $(6)_4$ sample points. This is because when no two faces are identical, the total number of possible outcomes is same as that of choosing 4 numbers from 1 to 6 (without replacement) for assigning them to the 4 throws. This gives the N for the conditional probability $P(A|B)$. Now let us count the number of cases (for the complementary event) when there is no ace, under constraint B. This is same as calculating N except now the numbers are to be chosen from 2 to 6, which yields the n of $P(A^c|B)$ as $(5)_4$. Thus $P(A|B) = 1 - \{(5)_4/(6)_4\}$. $\bigtriangledown$

A wary reader should ponder about the validity of the complementation law in the context of the conditional probability, which has been used in **Example 2.33** above. This point merits some discussion which will also help better understand the notion of conditional probability. All the probability laws discussed so far and the last one that will be presented shortly, are also valid under the conditional set-up. This is because according to ***Definition 2.4***, conditional probability is same as the "usual" probability, except that the calculation is done in a restricted sample space. Restricting the sample space to some set $B \subseteq \Omega$ instead of the original Ω might change the numerical value, but it does not alter the mathematical properties and characteristics of the intrinsic notion of probability. As a matter of fact all probabilities are conditional probabilities, and thus all the probability laws are equally applicable to the conditional probabilities as well. As stated in the first paragraph of this sub-section, probability of an event depends on one's state of information. In case of the "usual" unconditional probability, this state of knowledge is contained in Ω, and thus all the probabilities we had calculated till §2.5.1 were essentially $P(A|\Omega)$, but since this was the case across the board we did not complicate matters by using the conditional probability notation. But now that we are generalizing this notion to $P(A|B)$ for any arbitrary $B \subseteq \Omega$ it is important to realize that the state of knowledge or the sample space might change, but

[8]We are using such a characterization instead of say something like $\{\{g, g\}, \{g, b\}, \{b, b\}\}$ for making all the outcomes equally likely. In this later characterization the second element $\{g, b\}$ has a probability of 0.5 and the other two 0.25 each, while in the former characterization all the four outcomes gg, gb, bg, and bb are equally likely each with a probability of 0.25.

the basic laws and properties of probability remain intact even in this generalized conditional set-up. In order to prove the complementary law for conditional probability for instance, all one need to do is replace $P(\cdot)$ by $P(\cdot|B)$ and the same proof essentially goes through with a little careful reasoning. In general all the laws for conditional probabilities can be formally proved with the help of the multiplication law which is taken up next.

Multiplication Law: $P(A \cap B) = P(A|B)P(B) = P(B|A)P(A)$.
Proof: We shall provide a proof of this for the case of finite Ω with the loose definition of conditional probability given in ***Definition 4***. Let Ω have N elements with the number of elements in A, B and $A \cap B$ being n_A, n_B and n_{AB} respectively. Then by (1), $P(A \cap B) = n_{AB}/N$, $P(A) = n_A/N$, $P(B) = n_B/N$, and together with ***Definition 4***, $P(A|B) = n_{AB}/n_B$ and $P(B|A) = n_{AB}/n_A$ and the result follows. $\bigtriangledown$

In many text books, conditional probability is defined in terms of the multiplication law *i.e.* $P(A|B)$ is defined as $P(A \cap B)/P(B)$ when $P(B) > 0$ and undefined otherwise. While this is a perfect mathematical definition, and there is no other option but to define conditional probability in this way for a more mathematically rigorous treatment of the concept, in this author's opinion, this approach of defining conditional probability often obscures its intrinsic meaning and confuses the beginners in the subject in its elementary usage for solving everyday problems with which these notes are mainly concerned with. The reason for this is, as shall be seen shortly in miscellaneous examples, in elementary applications, one typically starts with an appraisal of conditional probabilities which are in turn used to figure out the joint probabilities $P(A \cap B)$ using the multiplication law and not the other way round. Thus if conditional probability is defined in terms of the joint probability, in such elementary applications, it puts the cart in front of the horse and in the process confuses the user. Therefore it is imperative that we first have an intuitive workable definition of conditional probability such as the one provided in ***Definition 2.4***, and then use this definition to prove the multiplication law. This approach not only facilitates conceptual understanding of conditional probabilities for its elementary everyday usage, for the cases where the conditional probability itself needs to be figured out[9], the multiplication law can be used as a result rather than the starting point of a definition. A couple of examples should help illustrate this point.

Example 2.34: An urn contains b_1 black balls and r_1 red balls. First a ball is drawn at random from this urn and its color is observed. If the color of the ball is black, the chosen ball is returned and an additional b_2 black balls are added to the urn. If the color of the ball is red then this ball and an additional r_2 many red balls are withdrawn from the urn. After this mechanism at the first step, a second ball is now drawn from the urn, and we are interested in the probability of this second ball being red. Let B_1 and R_1 respectively denote the event that the first ball chosen is black and red, and R_2 denote the event of interest, "the second ball chosen is red". Then

$P(R_2)$

[9] As for instance in **Example 2.33**, the interest was directly in $P(A|B)$. There we computed this probability from definition and got a counter-intuitive answer, and thus it might be illustrative to note that, using the multiplication law the answer is again $P(A|B) = P(A \cap B)/P(B) = 0.25/0.75 = 1/3$, irrespective of the kind of characterization being used to represent Ω.

$$
\begin{aligned}
&= P(B_1 \cap R_2) + P(R_1 \cap R_2) \text{ (since } R_2 = (B_1 \cap R_2) \cup (R_1 \cap R_2) \text{ and} \\
&\qquad\qquad (B_1 \cap R_2) \cap (R_1 \cap R_2) = \phi) \\
&= P(R_2|B_1)P(B_1) + P(R_2|R_1)P(R_1) \text{ (by multiplication law)} \\
&= \frac{b_1}{b_1+r_1}\frac{r_1}{b_1+b_2+r_1} + \frac{r_1}{b_1+r_1}\frac{r_1-r_2-1}{b_1+r_1-r_2-1} \qquad \bigtriangledown
\end{aligned}
$$

Example 2.35: Three men throw their hats and then randomly chooses one. We are interested in the probability that none of the men gets his own hat back. Probability of the complementary event, "at least one match" has already been worked out for the general case of n hats in **Example 30** and for $n = 3$ the answer to the question asked here is thus $1 - \left(1 - \frac{1}{2!} + \frac{1}{3!}\right) = \frac{1}{3}$. However here we shall see how conditional probabilities are used in figuring out the probability of event intersections to answer the question. As in **Example 30**, let A_k denote the event the k-th man got his hat back, $k = 1, 2, 3$. Then we are interested in the probability of the event $A_1^c \cap A_2^c \cap A_3^c$, which is same as $1 - P(A_1 \cup A_2 \cup A_3)$, and $P(A_1 \cup A_2 \cup A_3)$ is computed using (2). By (2),

$$
P(A_1 \cup A_2 \cup A_3) = P(A_1) + P(A_2) + P(A_3) - P(A_1 \cap A_2) - P(A_2 \cap A_3) - P(A_3 \cap A_1) + P(A_1 \cap A_2 \cap A_3). \tag{3}
$$

Obviously $P(A_k) = \frac{1}{3}\ \forall k = 1, 2, 3$, and $P(A_k \cap A_l)$ for $k \neq l$ is computed using the multiplication law as follows. $P(A_k \cap A_l) = P(A_k|A_l)P(A_l) = \frac{1}{2} \cdot \frac{1}{3}$, because given that the l-th man got his own hat back, the k-th man has two hats to choose from of which one is his own and thus $P(A_k|A_l) = \frac{1}{2}$. Again by the multiplication law $P(A_1 \cap A_2 \cap A_3) = P(A_1|A_2 \cap A_3)P(A_2 \cap A_3)$, and with $P(A_2 \cap A_3) = \frac{1}{6}$ as just shown above, we only need to figure out $P(A_1|A_2 \cap A_3)$. In words this requires the probability that the first man gets his own hat back, given that the other two got theirs, which is obviously 1, because in this case the first man has only one hat to choose from which is his own. Thus we get that $P(A_1 \cap A_2 \cap A_3) = \frac{1}{6}$, and after plugging in the required probability figures in equation (3), we get $P(A_1 \cup A_2 \cup A_3) = 3 \times \frac{1}{3} - 3 \times \frac{1}{6} + \frac{1}{6} = \frac{2}{3}$ and the probability of the event of interest as $\frac{1}{3}$. $\bigtriangledown$

As we saw in the above two examples, a large class of practical application problems require probabilities of intersections of events, which are typically worked out using the multiplication law with the required conditional probability values logically evaluated by implicit or explicit appeal to ***Definition 2.4***. The multiplication law is also sometimes used from the other direction for evaluating conditional probabilities, in which case there is no problem in viewing it as the definition of conditional probability, but logical problem persists for the former cases. A similar both ways application of a definition is used in practical applications for a closely related concept called **Statistical Independence** which is presented next, before looking at an array of examples applying all these concepts and laws.

2.5.2 Statistical Independence

We use the term "independent" in our everyday language to indicate two events having no effect on each other. For example one might say that the events stock market going up

tomorrow and rain today are independent, or wearing glasses and acing the Statistics course are independent. On the other hand events like raining and your vehicle starting up in the first crank, or getting an A in Statistics and an A in Finance might not be independent. Thus all of us use and have an intuitive understanding of what two events being independent mean. Here we shall formally study what independence means from a probabilistic point of view. As usual we start with the definition of independence.

Definition 2.5: Two events A and B are said to be **statistically** or **stochastically independent** (or simply **independent** in these notes) if $P(A|B) = P(A)$.

Before proceeding any further let us first try to understand why is independence defined in the above manner. According to ***Definition 2.5***, if the chance of occurrence of A remains unaltered with the additional information that B has already happened, then the events A and B are called independent. This makes a lot of intuitive sense because otherwise, if the knowledge of occurrence of B makes it either more or less likely for A to happen, then B is somehow influencing A and thus they should not be called independent in the usual sense of the word. While this definition is intuitively very appealing, an alternative, operationally slightly easier but equivalent result for independence of two events is as follows.

Proposition 2.1: Two events A and B are independent if and only if $P(A \cap B) = P(A)P(B)$.

The equivalence of ***Definition 2.5*** and ***Proposition 2.1*** follows in one step from the multiplication law, which also goes on to show that $P(A|B) = P(A) \Leftrightarrow P(B|A) = P(B)$, as one would expect in case A and B are independent. That is in the intuitive explanation of ***Definition 5*** or the definition itself, the role of A and B should be interchangeable and this shows that it is indeed so.

Just as in the case of multiplication law, ***Definition 2.5/Proposition 2.1*** is used both ways. By that it is meant that, often from the very structure of the problem independence is assumed, like for example it might be very reasonable to assume that the outcomes of two successive tosses of a coin are independent, and then this structural independence is used to compute the probabilities of joint events using ***Proposition 2.1***, like if for a given coin $P(H) = 0.6$, the probability of obtaining HH in two successive tosses of this coin is computed as $0.6 \times 0.6 = 0.36$. On the other hand many times there may not be any *apriori* reason to assume independence, and whether two events are independent or not is verified through ***Definition 2.5/Proposition 2.1***. These uses of independence are illustrated in the following examples.

Example 2.36: A card is drawn at random from a standard deck of 52 playing cards. Let the event A be, "the card drawn is an Ace", and the event B be, "the card drawn is a Spade". Since the four Aces are equally distributed across the four suits it is intuitively quite obvious that these two events must be independent. A formal check through ***Definition 5*** yields that $P(A) = 4/52 = 1/13$, while $P(A|B) = 1/13$ because given that we know that the card drawn is a Spade, our sample space gets reduced to that of the 13 spade suit cards in the deck, only one of which is an Ace, and thus $P(A|B) = 1/13$, showing that A and B are independent. $\bigtriangledown$

Example 2.37: Consider choosing one of the 720 permutations of the six letters, a, b, c,

d, e and f at random. Let A be the event that, "a precedes b" and B be the event, "c precedes d". The number of outcomes favorable to A equals $\binom{6}{2} \times 4!$ (choose any 2 of the 6 positions, place a in the lower and b in the higher ranking positions and then allow for all the 4! possibilities for the remaining 4 letters to occupy the remaining 4 places), similarly the number of outcomes favorable to B equals $\binom{6}{2} \times 4!$, and the number of outcomes favorable to $A \cap B$ equals $\binom{6}{2} \times \binom{4}{2} \times 2!$ (first choose the positions of a and b in $\binom{6}{2}$ ways, then choose the positions of c and d in $\binom{4}{2}$ ways from the remaining 4 positions and finally let e and f occupy the two remaining positions in 2! ways). Thus $P(A) = P(B) = 15 \times 24/720 = 1/2$ and $P(A \cap B) = 15 \times 6 \times 2/720 = 1/4$ and hence $P(A \cap B) = P(A)P(B)$ showing that they are independent[10]. $\bigtriangledown$

Example 2.38: Consider the experiment of rolling a white and a red fair die simultaneously. Let A be the event, "the white dice turned 4" and B be the event, "the sum of the faces equal 9". Then $P(A) = 1/6$ while $P(A|B) = 1/4$ showing that these two events are not independent. The intuitive reason behind dependence between A and B is as follows. If we already know that B has occurred then that precludes the result of the roll of the white dice from being an 1 or a 2 and thus increasing the chance of obtaining a 4 compared to the case when we have no information as to the occurrence of B. However if C denotes the event, "the sum of the faces equal 7" then this knowledge does not preclude any outcome of the white dice and thus A and C must be independent as is easily verified from $P(A \cap C) = \frac{1}{36} = \frac{1}{6} \cdot \frac{1}{6} = P(A)P(C)$. $\bigtriangledown$

Example 2.39: Statistical independence however may not always be intuitively obvious as the above examples might tend to suggest. Consider families with three children so that $\Omega = \{ggg, ggb, gbg, bgg, bbg, bgb, gbb, bbb\}$, where g stands for a girl and b stands for a boy. Now consider the events A, "the family has children of both genders" and B, "the family has at most one girl child". Then $P(A) = 6/8 = 3/4$ and $P(A|B)$ also equals $3/4$ because $B = \{bbg, bgb, gbb, bbb\}$, and in this restricted sample space A can happen for three of the outcomes bbg, bgb and gbb. Thus these two events are independent. However the events A and B are not independent for families with 2 children or 4 children for instance. $\bigtriangledown$

Example 2.40: In a similar vein, in a class with 4 Female Ph.D., 6 Female MBA, and 6 Male Ph.D. students, gender and degree would be independent if and only if there are exactly 9 Male MBA students. It is just a numerical fact and there is no intuitive reason behind this. $\bigtriangledown$

[10] Actually the combinatorial arguments are not needed to see that $P(A) = P(B) = 1/2$ and $P(A \cap B) = 1/4$. This is because in any of the permutations either a will precede b or b will precede a and they are equally likely because all possible permutations are being considered. With similar reasoning it can be seen the $P(B) = 1/2$. As far as the simultaneous positioning of a & b and c & d are concerned, there are four possibilities with each one being as likely as the other. Thus the event $A \cap B$, "a precedes b and c precedes d" has probability 1/4. This reasoning like the previous example makes it intuitively obvious why A and B should be independent.

Example 2.41: We close this subsection after providing an example of how ***Proposition 1*** may be used the other way round *i.e.* how it can be used to solve problems with assumed structural independence. Consider firing a flying target down by simultaneously using a surface-to-air missile and an air-to-air missile. Since the fighter on the ground, firing surface-to-air missile, and the airborne pilot firing air-to-air missile are physically acting independently of each other it may be reasonable to assume that the events of either one succeeding in firing the flying target down are statistically independent of one another. Now suppose the chance of the ground fighter succeeding is 0.95 and the chance of the airborne pilot succeeding is 0.99. We are interested in finding the probability of succeeding in firing the flying target down. If A denotes the event, "ground fighter succeeds" and B, "airborne pilot succeeds", then according to the above information, $P(A) = 0.95$, $P(B) = 0.99$) and A and B are independent, and we are to find $P(A \cup B)$. By addition law this equals 0.95+0.99-$P(A \cap B)$, and by ***Proposition 1***, $P(A \cap B) = P(A)P(B) = 0.95 \times 0.99 = 0.9405$ so that the probability of succeeding in firing the flying target down equals 0.95+0.99-0.9405=0.9995. $\bigtriangledown$

2.5.3 Bayes' Theorem

We shall now start looking at applications of different Probability laws that we have learned, some flavor of which has already been provided in a couple of examples above. By that it is meant that for instance, in **Examples 2.35** and **2.41** both multiplication and addition laws have been used in solving them. Likewise most real life problems require systematic analysis and then application of the appropriate law in solving them. Among these there is a class of problems which occur recurringly in applications. These class of problems require reevaluation or upgradation of probabilities of events when additional information is acquired. Actually in a nut-shell the entire business of statistical analysis, in one of the contemporary viewpoints, is viewed as above *i.e.* upgradation of probabilities in light of the collected data.

These class of problems are solved using **Bayes' Theorem**. Viewed as an off-shoot of the Probability laws, the theorem helps solve only one particular type of "application of probability law" problems. However because of its central role in the so-called Bayesian Statistics, this theorem requires special attention and a lot of importance is attached to this theorem in elementary probability theory. The Theorem goes as follows.

Bayes' Theorem: Let $A_1, \ldots, A_n$ denote n **mutually exclusive** and **exhaustive** states of nature *i.e.* $A_i \cap A_j = \phi$ for $i \neq j$ (mutually exclusive[11]) and $\cup_{i=1}^{n} A_i = \Omega$ (exhaustive). Suppose one starts with one's prior belief about the states of nature expressed in terms of

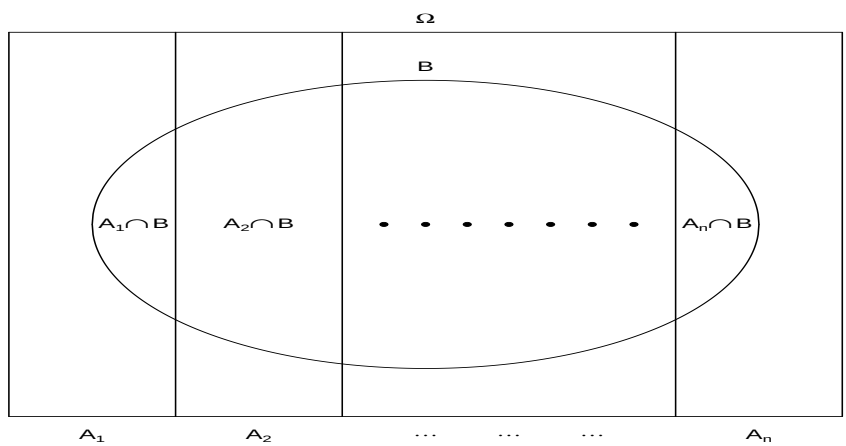
[11]Students tend to get confused between the notion of mutually exclusive and independent events. A and B mutually exclusive $\Leftrightarrow A \cap B = \phi$ while A and B independent $\Leftrightarrow P(A \cap B) = P(A)P(B)$. Thus if two events are mutually exclusive, they cannot be independent unless one of them is ϕ. Similarly if two events are independent they cannot be mutually exclusive unless one of them is ϕ. This should be intuitively obvious because if two events are mutually exclusive then they cannot happen simultaneously and thus if we know that one of them has happened then the other one cannot happen, and thus they cannot be independent. For example, when a card is drawn at random from an usual deck of 52 playing cards, then its suits or denominations are mutually exclusive - the card drawn cannot simultaneously be a Spade and a Club or an

the probabilities of A_i's, called the *apriori* probabilities. That is suppose someone believes that the probability that A_i will occur is π_i, $i = 1, \ldots, n$ so that $\pi_i \geq 0$ and $\sum_{i=1}^{n} \pi_i = 1$. Now suppose one collects some data which is expressed as the fact, "event B has occurred". Also suppose one has a statistical model which allows one to evaluate the chance of occurrence of the data B for each of the n alternative scenarios of states of nature $A_1, \ldots, A_n$ given by $P(B|A_1), \ldots, P(B|A_n)$. Given these and the fact that "event B has occurred", one upgrades one's belief about the n states of nature $A_1, \ldots, A_n$ from their *prior* probabilities $\pi_1, \ldots, \pi_n$ to their respective *posterior* probabilities $P(A_1|B), \ldots, P(A_n|B)$ as follows:

$$\text{For } i = 1, \ldots, n, \quad P(A_i|B) = \frac{\pi_i P(B|A_i)}{\sum_{j=1}^{n} \pi_j P(B|A_j)}. \tag{4}$$

<u>Proof:</u> The Venn diagram in Figure 3, where the n mutually exclusive and exhaustive states of nature have been represented by n non-overlapping vertical rectangles spanning the entire sample space Ω and the data B by the oval, will facilitate understanding the steps of the proof.

Figure 3: Venn Diagram for Bayes' Theorem



$$\begin{aligned}
&P(A_i|B) \\
&= \frac{P(A_i \cap B)}{P(B)} \text{ (by the multiplication law)} \\
&= \frac{P(B|A_i)P(A_i)}{P(B)} \text{ (again by the multiplication law)} \\
&= \frac{\pi_i P(B|A_i)}{P\left(\cup_{j=1}^{n}[A_j \cap B]\right)} \text{ (as } P(A_i) = \pi_i \text{ and } B = \cup_{j=1}^{n}[A_j \cap B] \text{ as } A_j\text{'s are exhaustive} \\
&\qquad\qquad\qquad\qquad \text{- see Figure 3)} \\
&= \frac{\pi_i P(B|A_i)}{\sum_{j=1}^{n} P\left(A_j \cap B\right)} \text{ (since} A_j \cap B\text{'s are mutually exclusive - again see Figure 3)}
\end{aligned}$$

Ace and a King; but the denomination and suit are independent of each other.

$$= \frac{\pi_i P(B|A_i)}{\sum_{j=1}^{n} P(B|A_j)\, P(A_j)} \text{ (by the multiplication law)}$$

$$= \frac{\pi_i P(B|A_i)}{\sum_{j=1}^{n} \pi_j P(B|A_j)} \text{ (as } P(A_j) = \pi_j) \qquad \bigtriangledown$$

Example 2.42: Suppose 75% of the students in a University lives on campus, and 80% of the students living off-campus and 50% of the students living on-campus owns a vehicle. What is the probability that a student owning a vehicle lives on campus? Here we have two mutually exclusive and exhaustive states of nature A_1 and A_2 denoting a student living "on" and "off" campus respectively with $P(A_1) = 0.75$ and $P(A_2) = 0.25$. Let B be the event of a student owning a vehicle. Then it is given that $P(B|A_1) = 0.5$ and $P(B|A_2) = 0.8$ and we are to find $P(A_1|B)$. By Bayes' theorem the required probability is given by $\frac{P(B|A_1)P(A_1)}{P(B|A_1)P(A_1)+P(B|A_2)P(A_2)} = \frac{0.5\times 0.75}{0.5\times 0.75+0.8\times 0.25} = 0.6522$. $\bigtriangledown$

Example 2.43: Suppose there are three chests and each chest has two drawers. One of the chests has a gold coin in each drawer, one of the other chests has a gold coin in one drawer and a silver coin in the other, and the remaining chest has a silver coin in each of its drawers. One of the chests is drawn at random and then one of its drawers is opened at random and a gold coin is found in that drawer. What is the probability that this chest contains a gold coin in its other drawer? Here there are three states of nature A_1, A_2 and A_3, where A_1 denotes the chest with gold coins in both of its drawers, A_2 denotes the chest with a gold and a silver coin in its two drawers and A_3 denotes the chest with silver coins in both of its drawers. Now let B denote the event that the coin found in the randomly opened drawer in the randomly chosen chest is gold. Then $P(A_1) = P(A_2) = P(A_3) = 1/3$ and $P(B|A_1) = 1$, $P(B|A_2) = 1/2$ and $P(B|A_3) = 0$, and we are to find $P(A_1|B)$. By Bayes' theorem this equals $\frac{1\times(1/3)}{1\times(1/3)+(1/2)\times(1/3)+0\times(1/3)} = \frac{2}{3}$. Note that the answer is <u>not</u> $\frac{1}{2}$ as some of you might have expected! $\bigtriangledown$

2.5.4 Examples

We finish this section (as well as this chapter on Elementary Probability Theory) by working out a few miscellaneous examples on Probability Laws. Unlike a discussion type format for the earlier examples we shall adopt a Problem-Solution format here for better clarity.

Example 2.44: A Sale is advertised in TV, Radio and Newspaper. The chance of a consumer watching it in TV is 40%, listening it in Radio is 15%, and reading it in Newspaper is 30%. Among those who have read it in Newspaper, 10% have heard it in Radio, 60% have seen it in TV, and 65% have heard it in at least one of the two media, Radio or TV. Among those who have not read it in Newspaper, the chance that they have not noticed it in at least one of the remaining two media, Radio or TV either is 90%. What is the probability that a consumer has noticed the advertisement of the Sale?

Solution: Let A, B and C respectively denote the events of a consumer noticing it in TV,

Radio and Newspaper. Then it is given that,

$$\begin{array}{lll} P(A) = 0.4 & P(B|C) = 0.1 & \\ P(B) = 0.15 & P(A|C) = 0.6 & P(A^c \cup B^c|C^c) = 0.9 \ , \\ P(C) = 0.3 & P(A \cup B|C) = 0.65 & \end{array}$$

and we are to find $P(A \cup B \cup C)$. We shall use (3) for this probability calculation. For the r.h.s of (3) $P(A)$, $P(B)$ and $P(C)$ are already given; $P(A|C) = 0.6$ & $P(C) = 0.3$ $\Rightarrow P(A \cap C) = 0.18$; and $P(B|C) = 0.1$ & $P(C) = 0.3 \Rightarrow P(B \cap C) = 0.03$, by the multiplication law. Since the addition law applies for the conditional probabilities as well, $0.65 = P(A \cup B|C) = P(A|C) + P(B|C) - P(A \cap B|C) = 0.6 + 0.1 - P(A \cap B|C) \Rightarrow P(A \cap B|C) = 0.05$ and thus by the multiplication law $P(A \cap B \cap C) = 0.015$. Thus the only term that remains to be figured out for applying (3) is $P(A \cap B)$. For this, notice that $A \cap B$ equals the mutually exclusive union of $(A \cap B \cap C)$ and $(A \cap B \cap C^c)$ so that its probability is sum of $P(A \cap B \cap C)$ and $P(A \cap B \cap C^c)$. With $P(A \cap B \cap C)$ already obtained, we only need to figure out $P(A \cap B \cap C^c)$. By the complementation law, $P(A \cap B|C^c) = 1 - P([A \cap B]^c|C^c) = 1 - P(A^c \cup B^c|C^c) = 1 - 0.9 = 0.1$ and $P(C^c) = 0.7$. Thus $P(A \cap B \cap C^c) = 0.07$, and therefore $P(A \cap B) = 0.015 + 0.07 = 0.085$. Now with all the elements in place we finally obtain $P(A \cup B \cup C) = 0.4 + 0.15 + 0.3 - 0.085 - 0.03 - 0.18 + 0.015 = 0.57$. $\bigtriangledown$

Example 2.45: By studying the past behavior of stocks A, B and C, owned by the same business group, it has been observed that the probability of B or C appreciating on any given day is 0.5. If A appreciates on a given day, the probability of B appreciating is 0.7, the probability of C appreciating is 0.6, and the probability of both B and C appreciating is 0.5. However if A does not appreciate on a given day, the probability of B appreciating is 0.2, the probability of C appreciating is 0.3, and the probability of both B and C appreciating is 0.1. What is the probability of all three of the stocks A, B and C appreciating on a given day?

Solution: Let A, B and C denote the events, stocks A, B and C appreciating on a given day, respectively. It is given that,

$$\begin{array}{lll} & P(B|A) = 0.7 & P(B|A^c) = 0.2 \\ P(B \cup C) = 0.5 & P(C|A) = 0.6 & P(C|A^c) = 0.3 \\ & P(B \cap C|A) = 0.5 & P(B \cap C|A^c) = 0.1 \end{array} \ ,$$

and we are to find $P(A \cap B \cap C)$. Since it is given that $P(B \cap C|A) = 0.5$ we shall be through if we can figure out $P(A)$. From the information given in the second column above, by addition law (for conditional probability) we have $P(B \cup C|A) = 0.8$ and similarly from the information in the third column, $P(B \cup C|A^c) = 0.4$. Let $P(A) = p$. Then since $B \cup C = [(B \cup C) \cap A] \cup [(B \cup C) \cap A^c]$ and the two sets in the square bracket are disjoint, by multiplication and complementation law we have, $0.5 = P(B \cup C) = P(B \cup C|A)P(A) + P(B \cup C|A^c)P(A^c) = 0.8p + 0.4(1-p)$, which after solving for p yields $P(A) = 0.25$, so that by multiplication law $P(A \cap B \cap C) = 0.125$ since $P(B \cap C|A) = 0.5$. $\bigtriangledown$

Example 2.46: A sleuth investigating the cause of the motor accident of Princess Diana believes that it's due to the chauffeur being intoxicated has probability 0.7, due to a camera flash on the chauffeur's eyes has probability 0.4 and these two events are independent. He

collects data on the causes of motor accidents and finds that statistically, the probability of a fatal motor accident is 0.8, if the chauffeur is intoxicated and no camera is flashed on his eyes; 0.3, if the chauffeur is not intoxicated and a camera is flashed on his eyes; 0.9, if the chauffeur is intoxicated and a camera is flashed on his eyes; and 0.1, if neither the chauffeur is intoxicated nor a camera is flashed on his eyes. Answer the following:

a. In light of the collected data, what should now be the sleuth's probabilities for the different causes of the accident?

b. Do the events of the chauffeur being intoxicated and a camera flash on his eyes still remain independent?

Solution (a): Let D denote the event, "the chauffeur was intoxicated"; F be the event, "a camera was flashed on the chauffeur's eye"; and B be the event of a fatal motor accident. Now define $A_1 = D \cap F^c$, $A_2 = D^c \cap F$, $A_3 = D \cap F$ and $A_4 = D^c \cap F^c$. Then it is given that D and F are independent and

$$\begin{array}{lll} P(D) = 0.7 & P(B|A_1) = 0.8 & P(B|A_3) = 0.9 \\ P(F) = 0.4 & P(B|A_2) = 0.3 & P(B|A_4) = 0.1 \end{array}.$$

We are to update the probabilities of the possible causes of the fatal motor accident of Princess D. There are 4 possible states of nature A_1, A_2, A_3 and A_4, and the statistical model probabilities of a fatal motor accident under these four distinct causes are given in terms of $P(B|A_i)$'s above. Thus given the fact that the accident did happen, we can update the probabilities of these causes or states of nature for the sleuth using Bayes' theorem. But this first requires the input of the sleuth's prior probabilities for the four distinct causes which are obtained as follows. $P(A_1) = P(D \cap F^c) = P(D)P(F^c) = 0.7 \times 0.6 = 0.42$, since D and F are independent[12], and similarly $P(A_2) = 0.3 \times 0.4 = 0.12$ and $P(A_3) = 0.7 \times 0.4 = 0.28$. Now by subtraction, $P(A_4) = 1 - 0.42 - 0.12 - 0.28 = 0.18 = 0.3 \times 0.6 = P(D^c)P(F^c)$[13], which gives us the prior probabilities of the sleuth.Thus the common denominator of (4) or $P(B) = 0.8 \times 0.42 + 0.3 \times 0.12 + 0.9 \times 0.28 + 0.1 \times 0.18 = 0.642$ and then by (4),

$$\begin{array}{ll} P(A_1|B) = 0.8 \times 0.42/0.642 = 0.5234 & P(A_3|B) = 0.9 \times 0.28/0.642 = 0.3925 \\ P(A_2|B) = 0.3 \times 0.12/0.642 = 0.0561 & P(A_4|B) = 0.1 \times 0.18/0.642 = 0.0280 \end{array}.$$

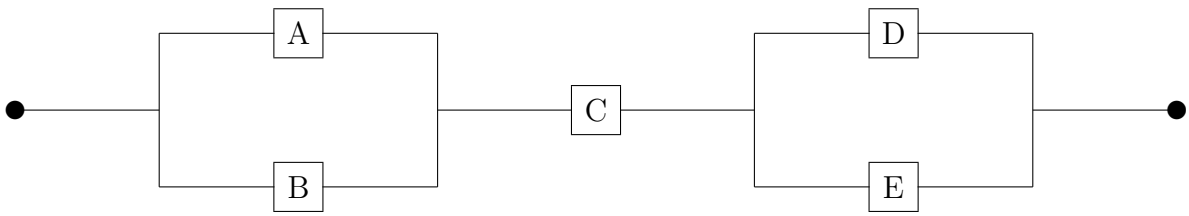
Hence to summarize, since $D = A_1 \cup A_3$ and $F = A_2 \cup A_3$, it may be stated that after collecting statistical data, *aposteriori* the sleuth must conclude that the chances of, the chauffeur being intoxicated was 91.59%, a camera flash was 44.86%, both were 5.61% and neither was 2.8%.

(b): Since we have just shown that *aposteriori* $P(D|B) = 0.9159$, $P(F|B) = 0.4486$ and $P(D \cap F|B) = 0.3925 \neq 0.4109 \approx P(D|B)P(F|B)$, the two events do not remain independent after observing the data. $\bigtriangledown$

Example 2.47: Consider a supply chain that starts with procurement by at least one of the two suppliers A or B, followed by a procurement by C and finally a procurement by at least one of D or E as illustrated in the following diagram:

[12] If A and B are independent then $P(A^c|B) = 1 - P(A|B) = 1 - P(A) = P(A^C)$ and thus A^c and B (and similarly A and B^c) are also independent.

[13] It is no surprise. In general if A and B are independent, so are A^c and B^c, which is proved as follows. $P(A^c \cap B^c) = P([A \cup B]^c) = 1 - P(A \cup B) = 1 - P(A) - P(B) - P(A)P(B) = (1 - P(A))(1 - P(B)) = P(A^c)P(B^c)$.



The item supplied by A or B depends on the weather condition and thus if B does not default, the probability of A defaulting is only 0.01. Marginally the probabilities of A and B defaulting are 0.05 and 0.1 respectively. C has defaulted 2% of the time in the past and behaves independently of others under all conditions. Both D and E behave independently of everybody else under all conditions and the marginal probabilities of their defaulting are 0.2 each. Answer the following:

a. What is the probability that the supply chain runs smoothly?

b. C is the most critical supplier in the sense that if C defaults the whole supply chain breaks down. Each one of the other suppliers has a back-up. Among these four suppliers with a back-up who is most critical and why?

c. If the supply chain breaks down, who is most likely to be responsible for it?

Solution (a): Let A, B, C, D and E respectively denote the events that suppliers A, B, C, D and E do not default *i.e.* able to procure their respective materials. Then the event that the supply chain runs smoothly, say S, may be expressed as the event $S = (A \cup B) \cap C \cap (D \cup E)$, so that the probability of this event of interest is given by,

$$
\begin{aligned}
&P(S) \\
&= P([A \cup B] \cap C \cap [D \cup E]) \\
&= P(A \cup B)P(C)P(D \cup E) \text{ (by independence)} \\
&= \{P(A) + P(B) - P(A \cap B)\} \times 0.98 \times \{P(D) + P(E) - P(D)P(E)\} \text{ (by addition and} \\
&\qquad \text{complementation law, given information, and independence of D and E)} \\
&= \{0.95 + 0.9 - (1 - 0.01) \times 0.9\} \times 0.98 \times \{0.8 + 0.8 - 0.8^2\} \text{ (since } P(A \cap B) \\
&\qquad = P(A|B)P(B) = (1 - P(A^c|B))P(B)) \\
&= 0.959 \times 0.98 \times 0.96 \\
&= 0.9022272
\end{aligned}
$$

(b): C is called most critical because $P(S|C^c) = 0$. While this is intuitively obvious from the block diagram, formally, $P(S|C^c) = P(\{[A \cup B] \cap C \cap [D \cup E]\} \cap C^c)/P(C^c) = P(\phi)/P(C^c) = 0$. Taking a cue from this, the criticality of supplier X may be judged by computing $P(S|$X defaults) for X=A, B, D and E and declaring the one to be most critical with smallest value of this probability.

$$
\begin{aligned}
&P(S|A^c) \\
&= P(\{[A \cup B] \cap C \cap [D \cup E]\} \cap A^c)/P(A^c) \text{ (by multiplication law)} \\
&= P([A^c \cap B] \cap C \cap [D \cup E])/0.05 \text{ (since} A \cap A^c = \phi) \\
&= P(A^c \cap B)P(C)P(D \cup E)/0.05 \text{ (by independence)} \\
&= (0.01 \times 0.9) \times 0.98 \times 0.96/0.05 \text{ (since } P(A^c \cap B) = P(A^c|B)P(B) \text{ and other numbers} \\
&\qquad \text{are as in } \textbf{(a)} \text{ above)} \\
&= 0.169344
\end{aligned}
$$

$$
\begin{aligned}
&P(S|B^c)\\
&= P(\{[A\cup B]\cap C\cap[D\cup E]\}\cap B^c)/P(B^c) \text{ (by multiplication law)}\\
&= P([A\cap B^c]\cap C\cap[D\cup E])/0.1 \text{ (since} B\cap B^c=\phi)\\
&= P(A\cap B^c)P(C)P(D\cup E)/0.1 \text{ (by independence)}\\
&= (0.95-0.99\times 0.9)\times 0.98\times 0.96/0.1 \text{ (since } P(A\cap B^c)=P(A)-P(A\cap B) \text{ and other}\\
&\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad \text{numbers as in } \textbf{(a)} \text{ above)}\\
&= 0.555072
\end{aligned}
$$

$$
\begin{aligned}
&P(S|D^c)=P(S|E^c) \text{ (by symmetry)}\\
&= P(\{[A\cup B]\cap C\cap[D\cup E]\}\cap E^c)/P(E^c) \text{ (by multiplication law)}\\
&= P([A\cup B]\cap C\cap[D\cap E^c])/0.2 \text{ (since} E\cap E^c=\phi)\\
&= P(A\cup B)P(C)P(D)P(E^c)/0.2 \text{ (by independence)}\\
&= 0.959\times 0.98\times 0.8\times 0.2/0.2\\
&= 0.751856
\end{aligned}
$$

Thus it may be concluded that, barring C, A is the most critical supplier followed by B and then D/E.

(c): Here we are to find $P(\text{Supplier X has defaulted}|S^c) = P(X^c|S^c)$ (say) for X=A, B, C, D and E and then point our finger to the most likely culprit based on these computed probabilities. Thus we need to find $P(A^c|S^c)$, $P(B^c|S^c)$, $P(C^c|S^c)$, $P(D^c|S^c)$ and $P(E^c|S^c)$. These probabilities are easily computed as follows. The default probabilities of the suppliers are given in the statement of the problem as

$$P(A^c)=0.05,\quad P(B^c)=0.1,\quad P(C^c)=0.02,\quad P(D^c)=0.2 \quad\text{and}\quad P(E^c)=0.2,$$

while $P(S^c|X^c)$ for $X=A$, B, C, D and E have been computed in part **(b)** of the problem (through the complementation law) as

$$P(S^c|A^c)=0.8307,\quad P(S^c|B^c)=0.4449,\quad P(S^c|C^c)=1 \quad\text{and}\quad P(S^c|D^c)=P(S^c|E^c)=0.2481,$$

and $P(S^c)$ has been computed (again through the complementation law) as 0.0978 in part **(a)** of the problem, so that by Bayes' Theorem, $P(X^c|S^c)$ may now be computed as $P(S^c|X^c)P(X^c)/P(S^c)$ for $X=A$, B, C, D and E as

$$
\begin{aligned}
&P(A^c|S^c)=0.8307\times 0.05/0.0978=0.4247 && P(B^c|S^c)=0.4449\times 0.1/0.0978=0.4549\\
&P(C^c|S^c)=1\times 0.02/0.0978=0.2045 \quad\text{and} && P(D^c|S^c)=P(E^c|S^c)=0.2481\times 0.2/0.0978\\
& && \qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad =0.5074.
\end{aligned}
$$

Thus if the Supply Chain breaks down, the most likely candidate is either D or E - both of them are equally likely of defaulting in case of the break down of the Supply Chain. $\triangledown$

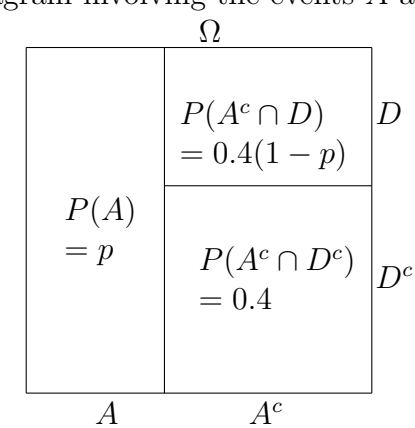
Example 2.48: By studying the past behavior of stocks A, B and C, owned by the same business group, it has been observed that the probability of none of the stocks appreciating on any given day is 0.4. If A does not appreciate on a given day, the probability of B appreciating is 0.2, the probability of C appreciating is 0.3, and the probability of both B

and C appreciating is 0.1. However if A appreciates on a given day, the probability of both B and C appreciating is 0.6. What is the probability of all three of the stocks A, B and C appreciating on a given day?

Solution: Let A, B and C denote the events, stocks A, B and C appreciating on a given day, respectively. It is given that,

$$P(A^c \cap B^c \cap C^c) = 0.4 \qquad \begin{array}{l} P(B|A^c) = 0.2 \\ P(C|A^c) = 0.3 \\ P(B \cap C|A^c) = 0.1 \end{array} \qquad P(B \cap C|A) = 0.6 \ ,$$

and we are to find $P(A \cap B \cap C)$. Just like in **Example 45** here also we shall be through if we can figure out $P(A)$ as we are given the value of $P(B \cap C|A)$. However here it is a little trickier to do so. Let $D = B \cup C$. Then from the information given in the second column above, by the addition law (for conditional probabilities), $P(D|A^c) = P(B|A^c) + P(C|A^c) - P(B \cap C|A^c) = 0.2 + 0.3 - 0.1 = 0.4$. Now note that since $D^c = (B \cup C)^c = B^c \cap C^c$ we are also given that $P(A^c \cap D^c) = 0.4$. Let $P(A) = p$. Then $P(A^c \cap D) = P(D|A^c)P(A^c) = 0.4(1-p)$. Now consider the following Venn diagram involving the events A and D:



Thus we have $p + 0.4(1-p) + 0.4 = P(\Omega) = 1$, solving which we get $P(A) = p = \frac{1}{3}$ and therefore $P(A \cap B \cap C) = P(B \cap C|A)P(A) = 0.6 \times \frac{1}{3} = 0.2$. $\bigtriangledown$

Example 2.49: Consider the famous Monty Hall problem. In a TV game-show, there are three closed doors and there is a prize behind one of these doors. A contestant in the show chooses a door at random, and then the host of the show, who knows the door with the prize behind it, (but probably pretends not to, in the interest of the show, dramatically) opens one of the two remaining doors, not chosen by the contestant, to show that the prize is not there. The contestant is now given a choice between sticking to the door originally chosen by her and switching her selection to the other closed door. The question is, "Should she switch?" and the answer somewhat surprisingly is YES! The solution to this problem is as follows.

Solution: Let A denote the event, "prize behind the door first chosen by the contestant" and B be the event, "contestant gets the prize by switching her choice of door". Then clearly $P(A) = \frac{1}{3}$ and if $P(B) > P(A)$ then it is better to switch the door because that improves the odds of winning the prize.

$P(B)$

$$
\begin{aligned}
&= P(B \cap A) + P(B \cap A^c) \text{ (because } B = (B \cap A) \cup (B \cap A^c) \text{ and } (B \cap A) \cap (B \cap A^c) = \phi) \\
&= P(B|A)P(A) + P(B|A^c)P(A^c) \text{ (by multiplication law)} \\
&= 0 \times \frac{1}{3} + 1 \times \frac{2}{3} \text{ (because if the door initially chosen by the contestant contains the prize} \\
&\quad \textit{i.e.} \text{ if it is known that } A \text{ has happened, then the chance of winning the prize by switch-} \\
&\quad \text{ing is 0 or } P(A|B) = 0\text{; and likewise if } A^c \text{ happens } \textit{i.e.} \text{ if the door originally chosen does} \\
&\quad \text{not contain the prize, then one is sure to win the prize by switching because the other} \\
&\quad \text{door not containing the prize has already been opened and thus } P(B|A^c) = 1) \\
&= 2/3.
\end{aligned}
$$

Therefore it is better to switch the door as it doubles the probability of winning the prize. This may seem counter-intuitive at first, because it appears that no matter which door is chosen by the contestant first - the one with the prize or one without it, the host can always open a door not containing the prize and thus the odds of winning by switching should remain the same as it was in the beginning for the switched door. But that is not the case because even intuitively now there is one less door and thus it should improve the odds (though note that the probability of interest is not 1/2 as one might intuitively guess with this later argument, because the probability sought is that of winning after switch, which is 2/3, and not that of winning after one of the doors is eliminated). $\bigtriangledown$

Problems

2.1. Consider the problem of distributing 3 balls in 3 cells. Assume that (I) both the balls and the cells are distinguishable.

a. Write down/enumerate the sample space.

b. What is the probability that exactly one of the cells is empty?

Answer the above under the assumption that (II) the balls are indistinguishable but the cells are and (III) both the balls the cells are indistinguishable. For answering **(b)** assume all the sample points in **(a)** are equally likely for each of above models I, II and III.

2.2. A toothbrush manufacturer claims that at least 40% of the dentists recommend their brand of toothbrush. In a random sample of 12 dentists 5 were found recommending the brand. In light of this data can the manufacturer's claim be validated?

2.3. In an office with 11 employees and one boss, a rumor about the boss has been started by one of the employees by telling it to another employee (excluding the boss) chosen at random, who in turn repeats it to a random third person and so on. At each consecutive step the recipient of the rumor is chosen at random from the remaining 10 persons in the office, which exclude the repeater and the person who told it to the repeater, but include the boss. Find the probability that the rumor will be circulated 5 times avoiding the boss.

2.4. An advertiser has given 10 placards to be put up around a departmental store, which has 6 different locations for putting up such campaigns. Imagine that each location has an unlimited capacity of holding placards. If the placards are assigned to the 6 locations

at random, what is the probability that each one of the 6 locations will hold at least one placard?

2.5. n items, among whom are A and B, are to be displayed on the shelf of a departmental store in a row. If all possible arrangements are equally likely, what is the probability that there will be exactly r items between A and B? Show that if the n items are displayed on a circular table forming a ring instead, and if all possible arrangements are again equally likely, the probability of having exactly r items between A and B in the clock-wise direction is free from r.

2.6. The personnel manager of a financial institution is to distribute 10 freshly recruited management trainees to one of its 4 zonal head-offices. If she assigns the trainees at random, what is the probability that each of the zonal head-offices receives at least one trainee?

2.7. 5 operators are to be assigned to operate 3 machines. If every machine is to get at least one operator, what is the probability that the first machine has two operators assigned to it?

2.8. In a factory 4 operators take turn in setting up a machine. An improper set up causes a break-down of the machine. Out of the 4 break-downs 3 occurred after operator A had set it up. Find the probability of occurrence of 3 or more break-downs (out of 4) due to operator A. So can the observed event be attributed to chance alone, or is it justifiable to say that operator A is worse than the others, so that he needs some extra training, for instance?

2.9. Among the starting offers of 4 fresh Engineering graduates and 5 fresh Management graduates it is observed that the top 4 offers belong to the Management graduates. Assume that there are no ties among the 9 offers. If the probability distributions of the starting offers of both the fresh Engineering and Management graduates were same, all possible arrangements of the offers would have been equally likely. Under such an assumption, what is the probability of observing 4 or more of the top offers belonging to the Management graduates? So, do the probability distributions of the starting offers of both the fresh Engineering and Management graduates appear to be same?

2.10. The board of directors of a private limited company have 15 members, out of whom 3 are members of the family of the major share-holder of the company. A 5 member committee is formed from the board of directors and 2 of the 3 family members happen to represent in the committee. Is there any strong evidence of nepotism? What would be your conclusion if all 3 of the family members are represented in the committee?

2.11. While evaluating the feasibility of undertaking a new project, D, the leader of a team of 4 programmers A, B, C and D, analyzes that only A has the skill to write the initial part of the code and (subjectively) assesses the probability of A being able to implement it successfully to be 0.9. For the remainder, she (D) alone can successfully write the code with a probability of 0.8, or divide the work between B and C so that after B finishes writing his part, which has a success probability of 0.7, C can take over and finish it off with a probability of 0.95. Assume that the events of any one of them being successful are mutually independent. What is the maximum probability of the project being successfully completed?

2.12. In the decision making process of a company, a resolution is taken if the President (P) approves it, or if both the Managing Director (MD) and the General Manager (GM) approve it. P's decision is independent of that of the MD and/or the GM. On the issue of a new purchase, the probability that P will approve it 0.6. If the MD approves of the purchase, which has probability 0.8, the chance that the GM will support the MD is 0.5. What is the probability that the purchase will be approved by the management?

2.13. An investor is speculating whether the value of a certain stock, he is holding, will go up further tomorrow, compared to its value today, for otherwise he can sell them for a profit today. His broker tells him that he has a strong personal feeling that its value is going to appreciate tomorrow, to which he is assigning a prior subjective probability of 0.8. However, from the past data on that particular stock the investor observes that, among the days the stock has appreciated, 20% of the time it has also appreciated on the previous days. On the other hand, among the days it has depreciated, its value has still gone up on 90% of the previous days. The stock has appreciated today. What is the probability that the stock will appreciate tomorrow?

2.14. While trying to develop strategies for launching a new product, three ideas, say A, B and C emerged from the marketing team. From his past experiences with similar products and business strategies the marketing Vice-President of the company envisages that B is twice, C is half, and at least one of A, B, or C is two and half times as likely to be successful as A alone. He also appraises that if B succeeds the chance of A succeeding is 0.2, if C succeeds the chance of A succeeding is 0.9, and the chance of all three succeeding is 0.1. If all the three strategies are possible to implement simultaneously and the success of strategy B is independent of the success of strategy C, what is the probability of at least one of the strategies being successful? Strategies A and B being nearly complementary in nature, the proponent of strategy A argues that while she quite agrees with all the remaining subjective appraisals of the Vice-President, she believes that if B succeeds the chance of A succeeding is 35% and not 20%. The Vice-President showed that, that leads to an inconsistency. What was the Vice-President's argument?

2.15. A mining company has 500 miners, 100 engineers and 50 management personnel. Among the miners 25% have no children and 30% have one child. Among the engineers 45% have no children and 35% have more than one child. Among the managers 20% have no children and 65% have more than one child. What is the probability that a randomly selected employee of the company has

a. no children?

b. has one child?

c. has more than one child?

2.16. The defect rates of machines A and B are 5% and 1% respectively. 50% of the products are manufactured using machine A. What is the probability that a defective product has been manufactured by machine A?

2.17. A departmental store specializing in men's apparel, sell Dress and Accessories. Accessories are further classified into Cloth (tie, handkerchief etc.) and Leather (shoe, belt etc.).

It is found that 80% of the purchases are Dress while 48% of the purchases are Accessories. Among the customers purchasing Dress, 20% purchase Cloth Accessories, and 18.75% purchase Leather Accessories. Among the customers who do not purchase Dress, 70% purchase Cloth Accessories, and 50% purchase Leather Accessories. The *Affinity* of Product A *to* Product B is defined as the conditional probability of purchase of A given a purchase of B.

a. Find the Product Affinity of Dress to Accessories.

b. Find the Product Affinity of Dress to Cloth Accessories.

c. Find the Product Affinity of Dress to Leather Accessories.

d. Find the Product Affinity of Accessories to Dress.

e. Show that as such the purchase of a Cloth Accessory and a Leather Accessory are not independent of each other, however they are independent conditional on a Dress purchase.

2.18. The probability that the first launch of a satellite by a company is successful is 0.75. The probability of a consequent successful launch, preceded by a successful one is 0.8, while preceded by a failed one is 0.9. What is the probability that the third launch by the company will be successful?

2.19. The probability that a vehicle passes by during any given second at a particular point on a road is p. A pedestrian can cross that particular point on the road if there is no car passing by for two consecutive seconds. Treating the seconds as indivisible time units, find the probability that a pedestrian has to wait for $k = 0, 1, 2, 3$ seconds.

2.20. Consider a communication network of 4 nodes where there is a direct link between any two nodes. The probability of a direct link between two nodes going down is 0.05 and each direct link behaves independently of one another. If two nodes can communicate as long as there is a link between them, what is the probability that two given nodes A and B can communicate with each other?

2.21. As promised to a team of 3 programmers, A, B and C, at least one of them is to be promoted after the successful completion of a Project. Both B and C will not be promoted simultaneously, and each one has 40% chance to get promoted. If B is promoted there is a 25% chance that A will also get a promotion. The chance of both A and C getting promoted simultaneously is 20%. Answer the following:

a. Find the probability of both A and B getting promoted simultaneously.

b. What is the probability of A getting a promotion?

c. Answer the same in **(b)**, if you have the additional information that "C is promoted".

d. If A gets a promotion, what is the probability that

i B also gets a promotion?
ii C also gets a promotion?

e. What is the probability of

i A alone getting a promotion?
ii B alone getting a promotion.
iii C alone getting a promotion

f. Find the probability of at least two of them getting a promotion.

g. Are the events of their getting promoted independent of each other? Discuss in detail.

2.22. A polygraph (lie-detector) test correctly indicates when a person is lying 95% of the time, while it lets 90% of the innocents free. The judge in a trial feels that there is about 70% chance that a certain witness is lying and ordered a polygraph test, which showed negative (i.e. the person is not lying). What is the judge's updated probability of the witness lying in view of the result of the polygraph test?

2.23. Given any three events A, B and $C \in \mathcal{A}$, the σ-field of events, show that the event, "Exact1y 2 of the events A, B and C have occurred" a1so belongs to $\mathcal{A}$.

2.24. Show that for any n events $A_1, \ldots, A_n \subseteq, \Omega$ $P\left(\bigcup_{i=1}^{n} A_i\right) \leq \sum_{i=1}^{n} P(A_i)$.

2.25. Let $\Omega = (0, 1]$, $\mathcal{A} = \sigma$-field generated by all finite unions of intervals of the form $\cup_{i=1}^{n}(a_i, b_i]$ where $0 < a_1 < b_1 < a_2 < b_2 \cdots < a_n < b_n \leq 1$ and the probability of a set of the form $A = \cup_{i=1}^{n}(a_i, b_i]$ is defined as $P(A) = \sum_{i=1}^{n}(b_i - a_i)$, with probability of all other sets in $\mathcal{A}$ are defined using limiting arguments. Show that

a. Any finite set $X = \{x_1, x_2, \ldots, x_k\}(\subseteq \Omega) \in \mathcal{A}$ where each $x_i \in (0, 1]$ for $i = 1, 2 \ldots k$, and

b. $P(X) = 0$.